The Square Root of the Igusa Partition Function

A Borcherds lift of the weak Jacobi form $\phi_{0,1}$, generalized Borcherds–Kac–Moody superalgebras, and the Igusa cusp form Δ_5

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Abstract

For $X = S \times E$,

$$Z_{\mathrm{OP}}^{X} = -\left(\chi_{10}^{\mathrm{OP}}\right)^{-1} = -4096\,\Delta_5^{-2}.$$

With Eichler–Zagier normalization $Z_{\mathrm{K3}} = 2\phi_{0,1}$, write

$$\phi_{0,1} = \sum f(n,l) q^n r^l.$$

Set

$$[\mathbb{U}_{n,l}] \in K_0(\mathrm{sVect}_{\mathrm{fd}}), \qquad \mathrm{sdim}\,\mathbb{U}_{n,l} = f(n,l), \qquad \mathcal{V}_{(n,l,m)} := \mathbb{U}_{nm,l}.$$

$$\mathcal{D}_X = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma=(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5, \qquad \mathcal{D}_X^2 = -4096/Z_{\mathrm{OP}}^{X}.$$

$$\langle \gamma, \gamma' \rangle_{\mathrm{ind}} = 2(nm' + n'm) - ll', \qquad \alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2},$$

$$(\alpha(\gamma), \alpha(\gamma')) = -2\langle \gamma, \gamma' \rangle_{\mathrm{ind}}, \qquad C_{\mathrm{Borch}} = \mathbb{R}_{\geq 0}\alpha(\Gamma_{\mathrm{act}}).$$

$$\mathcal{A}_{\mathrm{den}} := \mathfrak{g}_{\Delta_5}, \qquad \mathrm{den}(\mathcal{A}_{\mathrm{den}}) = 64^{-1}\Delta_5(2Z).$$

For $\gamma = (n,l,m) \in \Gamma_{\mathrm{act}}$,

$$\mathrm{smult}(\alpha(\gamma)) = f(nm,l).$$

The microscopic comparison problem is

$$\mathrm{Prim}_{\mathrm{prot}}\,\mathcal{P}_{X,\sigma,o}^{\mathrm{Hall}} \overset{?}{\cong} \mathcal{A}_{\mathrm{den}}.$$

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1 The duality principle

$$\mathcal{D}_X = \Delta_5 \in M_5(\mathbf{Sp}_4(\mathbb{Z}), \nu_{\Delta_5}),$$

$$\mathcal{D}_X|_{\mathfrak{c}_\infty} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

For

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix},$$

put

$$q = e^{2\pi i z_1}, \qquad r = e^{2\pi i z_2}, \qquad s = e^{2\pi i z_3}.$$

For

$$T(n,l,m) = \begin{pmatrix} n & l/2 \\ l/2 & m \end{pmatrix}, \qquad n, l, m \in \mathbb{Z},$$

put

$$(T(n,l,m), Z)_{\mathrm{tr}} = \mathrm{tr}(T(n,l,m)Z) = n z_1 + l z_2 + m z_3.$$

Then

$$x^{(n,l,m)} = \exp(2\pi i (T(n,l,m), Z)_{\mathrm{tr}}) = q^n r^l s^m.$$

Definition 1.1 *(Index-lattice notation).*

$$\Gamma_{\mathrm{gram}} := \Gamma_{\mathrm{ind}} := \mathbb{Z}^3, \qquad \Gamma_{\mathrm{act}} \subset \Gamma_{\mathrm{eff}} \subset \Gamma_{\mathrm{gram}}.$$

Here Γ_{eff} is the cusp-positive semigroup and Γ_{act} is the non-zero Jacobi-support locus. No microscopic charge lattice is constructed from this notation.

PROPOSITION 1.2 *(Charge matrix convention).* Assume a protected charge grading with primitive electric-magnetic pair Q, P in an even lattice. Set

$$n = \frac{(Q,Q)}{2}, \qquad l = (Q,P), \qquad m = \frac{(P,P)}{2}.$$

Then

$$\begin{pmatrix} (Q,Q) & (Q,P) \\ (P,Q) & (P,P) \end{pmatrix} = 2T(n,l,m),$$

and the invariant discriminant is

$$\det(2T(n,l,m)) = 4nm - l^2.$$

The genus-two chemical potential Z pairs with this same matrix:

$$\exp(2\pi i\ \mathrm{tr}(TZ)) = q^n r^l s^m.$$

Thus the variables in the Borcherds product record the Gram invariants of this protected charge grading.

Proof. The charge lattice is even, hence $(Q,Q)/2$ and $(P,P)/2$ are integral and $(Q,P) \in \mathbb{Z}$. The displayed matrix identity is immediate from the definitions. Its determinant is $4nm - l^2$. The last equality is the definition of the Fourier character attached to the chemical potential Z. This is the Harvey–Moore and Dijkgraaf–Verlinde–Verlinde charge grading used by the BPS product [21], [11]. □

Definition 1.3 *(Gram-index pairing).* For

$$\gamma = (n,l,m), \qquad \gamma' = (n',l',m'),$$

define

$$\langle \gamma, \gamma' \rangle_{\mathrm{ind}} = 2(nm' + n'm) - ll'.$$

The physical discriminant is

$$\Delta(\gamma) = \langle \gamma, \gamma \rangle_{\mathrm{ind}} = 4nm - l^2.$$

Definition 1.4 *(Cusp polarization).* The standard cusp polarization is the ordered limit

$$0 < |s| \ll |q| \ll |r|^{-1} \ll 1.$$

It gives exactly the effective semigroup

$$\Gamma_{\mathrm{eff}} = \{(n,l,m) \mid m > 0,\ n \geq 0,\ l \in \mathbb{Z}\} \cup \{(n,l,0) \mid n > 0,\ l \in \mathbb{Z}\} \cup \{(0,l,0) \mid l < 0\}.$$

Remark 1.5 *(Chamber transport).* The pairing $\langle\ ,\ \rangle_{\mathrm{ind}}$ is the pairing on Gram invariants. The order on Γ_{eff} is a Fock polarization at one cusp. A duality element transports this polarization to the corresponding product chamber at the transported cusp; it does not preserve the displayed inequalities inside the same coordinate chart.

2 The one-particle index and its determinant

The Gram-index lattice is

$$\Gamma_{\mathrm{ind}} = \mathbb{Z}^3, \qquad \gamma = (n,l,m), \qquad x^\gamma = q^n r^l s^m.$$

The cusp polarization of Section 1 fixes

$$\gamma > 0 \iff \gamma \in \Gamma_{\mathrm{eff}}.$$

Definition 2.1 *(Ramond-sector K3 index and its half).* Let $\mathcal{H}_{\mathrm{RR}}(S)$ be the Ramond–Ramond Hilbert space of a unitary $\mathcal{N} = (4,4)$ sigma model on a K3 surface S. Its central charges are

$$c = \overline{c} = 6.$$

Choose a right-moving Ramond supercharge Q_R with

$$Q_R^2 = 0, \qquad \{Q_R, Q_R^\dagger\} = 2\left(\overline{L}_0 - \frac{\overline{c}}{24}\right).$$

The K3 elliptic genus is the protected supertrace

$$Z_{\mathrm{K3}}(\tau, z) = \mathrm{Tr}_{\mathcal{H}_{\mathrm{RR}}(S)}(-1)^{F_L+F_R} q^{L_0 - c/24} \overline{q}^{\overline{L}_0 - \overline{c}/24} r^{J_0}.$$

The BPS space in degree (n, l) is

$$\mathcal{H}^{\mathrm{BPS}}_{n,l}(S) := H^\bullet\big(\{\psi \mid (L_0 - c/24)\psi = n\psi,\ J_0\psi = l\psi\}, Q_R\big).$$

LEMMA 2.2 *(Right-moving localization).* Then

$$Z_{\mathrm{K3}}(\tau, z) = \sum_{n \geq 0,\ l \in \mathbb{Z}} \mathrm{sdim}\, \mathcal{H}^{\mathrm{BPS}}_{n,l}(S) q^n r^l.$$

Proof. If $\overline{L}_0 - \overline{c}/24 > 0$, the displayed anticommutator pairs states with opposite $(-1)^{F_R}$ parity. Their supertrace cancels. The unpaired states are the Q_R-harmonic representatives, hence the graded superdimension of Q_R-cohomology gives the elliptic genus. □

Definition 2.3 *(Half-K3 index object).* Eichler–Zagier normalization [12]:

$$Z_{\mathrm{K3}}(\tau, z) = 2\phi_{0,1}(\tau, z),$$

$$\phi_{0,1}(\tau, z) = \sum_{n \geq 0,\ l \in \mathbb{Z}} f(n, l) q^n r^l.$$

Thus

$$\mathrm{sdim}\, \mathcal{H}^{\mathrm{BPS}}_{n,l}(S) = 2f(n, l).$$

The class

$$\mathbb{U}_{n,l} \in K_0(\mathrm{sVect}_{\mathrm{fd}}), \qquad \mathrm{sdim}\, \mathbb{U}_{n,l} = f(n, l),$$

is the normalized half-index object used by the Borcherds lift. It is not asserted to be a canonical half of $\mathcal{H}^{\mathrm{BPS}}_{n,l}(S)$ as a Hilbert space.

$$\phi_{0,1} = r^{-1} + 10 + r + O(q), \qquad \frac{f(0,0)}{2} = 5.$$

LEMMA 2.4 *(Fock trace and superdeterminant).* Let

$$V = \bigoplus_{\lambda \in P} V_\lambda$$

be a P-graded super vector space with finite-dimensional homogeneous pieces and let $a_\lambda = \operatorname{sdim} V_\lambda$. Put

$$\mathcal{F}(V) = \operatorname{Sym}(V_{\bar{0}}) \otimes \bigwedge(V_{\bar{1}}).$$

As a formal character,

$$\operatorname{sch} \mathcal{F}(V) := \sum_{\mu \in P} \operatorname{sdim} \mathcal{F}(V)_\mu \, x^\mu = \prod_{\lambda \in P} (1 - x^\lambda)^{-a_\lambda}.$$

The inverse character is the superdeterminant

$$\operatorname{sdet}_V(1 - x) = \prod_{\lambda \in P} (1 - x^\lambda)^{a_\lambda}.$$

Proof.

$$V_\lambda = \mathbb{C}_{\bar{0}} : \qquad \operatorname{sch} \operatorname{Sym}(V_\lambda) = (1 - x^\lambda)^{-1}.$$

$$V_\lambda = \mathbb{C}_{\bar{1}} : \qquad \operatorname{sch} \bigwedge(V_\lambda) = 1 - x^\lambda.$$

Tensor product over a homogeneous basis proves the product. □

$$x^{(n,l,m)} = q^n r^l s^m.$$

$$\mathcal{V} = \bigoplus_{\gamma \in \Gamma_{\text{eff}}} \mathcal{V}_\gamma, \qquad \mathcal{V}_{(n,l,m)} = \mathcal{V}_{n,l,m} := \mathbb{U}_{nm,l}, \qquad \operatorname{sdim} \mathcal{V}_{n,l,m} = f(nm, l).$$

The $m > 0$ sector is the K_0-level virtual half of the DMVV product [10], obtained by replacing $Z_{\mathrm{K}_3} = 2\phi_{0,1}$ by $\phi_{0,1}$. The normalized Fock character in the Borcherds chamber is

$$\operatorname{sch} \mathcal{F}(\mathcal{V}) = \prod_{\gamma \in \Gamma_{\text{eff}}} (1 - x^\gamma)^{-\operatorname{sdim} \mathcal{V}_\gamma} = \prod_{(n,l,m) \in \Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{-f(nm,l)}.$$

$$\operatorname{sdet}_{\mathcal{V}}(1 - x) = \prod_{\gamma \in \Gamma_{\text{eff}}} (1 - x^\gamma)^{\operatorname{sdim} \mathcal{V}_\gamma} = \prod_{(n,l,m) \in \Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

The sector $m > 0$ is the normalized half of the second-quantized DMVV index. The terms with $m = 0$ are cusp and Weyl-chamber boundary factors of the Borcherds product; the determinant reads only their signed indices $f(0, l)$.

$$v_0(Z) = 64 q^{1/2} r^{1/2} s^{1/2}.$$

$$v_0 \notin \bigoplus_{\gamma \in \Gamma_{\text{eff}}} \mathcal{V}_\gamma.$$

Definition 2.5 (Virtual protected determinant).

$$\mathcal{D}_X(Z) = v_0(Z) \operatorname{sdet}_{\mathcal{V}}(1 - x) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

$$\mathcal{Z}^{\text{vir}}_{\text{BPS}}(Z) = \mathcal{D}_X(Z)^{-1} = \frac{1}{64} q^{-1/2} r^{-1/2} s^{-1/2} \prod_{(n,l,m) \in \Gamma_{\text{eff}}} (1 - q^n r^l s^m)^{-f(nm,l)}.$$

PROPOSITION 2.6.

$$[q^{1/2}r^{1/2}s^{1/2}]\,\mathcal{D}_X = 64,$$
$$\mathcal{Z}^{\mathrm{vir}}_{\mathrm{BPS}} = \frac{1}{64}q^{-1/2}r^{-1/2}s^{-1/2}\,\mathrm{sch}\,\mathcal{F}(\mathcal{V}),$$
$$\mathcal{D}_X = 64q^{1/2}r^{1/2}s^{1/2}\,\mathrm{sdet}_{\mathcal{V}}(1-x).$$

Proof.

$$\mathrm{sch}\,\mathcal{F}(\mathcal{V}) = \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}}(1-q^n r^l s^m)^{-f(nm,l)}, \qquad \mathrm{sdet}_{\mathcal{V}}(1-x) = \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}}(1-q^n r^l s^m)^{f(nm,l)}.$$

$$v_0(Z)\,\mathrm{sdet}_{\mathcal{V}}(1-x) = \mathcal{D}_X, \qquad v_0(Z)^{-1}\,\mathrm{sch}\,\mathcal{F}(\mathcal{V}) = \mathcal{Z}^{\mathrm{vir}}_{\mathrm{BPS}}.$$

□

THEOREM 2.7 *(Virtuality of the square root).* The displayed construction produces a class

$$[\mathcal{V}] \in K_0(\Gamma_{\mathrm{eff}}\text{-graded finite-dimensional super vector spaces})$$

together with its Fock determinant. By itself this determinant does not construct a Hilbert space, a vertex algebra, or a factorization algebra whose character is $\mathcal{Z}^{\mathrm{vir}}_{\mathrm{BPS}}$.

All denominator statements use only the equalities

$$\mathrm{sdim}\,\mathcal{V}_{(n,l,m)} = f(nm,l), \qquad \mathcal{D}_X = v_0\,\mathrm{sdet}_{\mathcal{V}}(1-x).$$

The denominator algebra and formal current envelope are constructed later from the Borcherds root datum, not from a canonical half of the K3 Hilbert space.

Proof. By definition, $\mathcal{V}_{(n,l,m)} = \mathbb{U}_{nm,l}$ is specified only through its class in $K_0(\mathrm{sVect}_{\mathrm{fd}})$ and through the integer $\mathrm{sdim}\,\mathbb{U}_{nm,l} = f(nm,l)$. The Fock trace and superdeterminant lemma depends only on these signed dimensions. No operation in the construction supplies composition maps, operator products, locality, or a Hilbert completion. □

Open Problem 2.8 *(Honest chiral square-root Hilbert lift).* Construct, if it exists, an honest chiral one-particle Hilbert space with a Γ_{eff}-graded super-vector-space decomposition

$$\mathcal{H}^{\mathrm{rp,ch}} = \bigoplus_{\gamma\in\Gamma_{\mathrm{eff}}}\mathcal{H}^{\mathrm{rp,ch}}_{\gamma}$$

with protected superdimensions

$$\mathrm{Tr}_{\mathcal{H}^{\mathrm{rp,ch}}_{(n,l,m)}}(-1)^F = f(nm,l).$$

The determinant construction uses only the virtual classes $\mathbb{U}_{n,l}$ and their signed dimensions. The algebraic object constructed from the same charge grading and the same signed multiplicities is the Borcherds denominator algebra.

3 The normalized cusp form

Freitag [13] and Van der Geer [34]: the even-weight scalar subring is

$$M_*^{\mathrm{even}}(\mathbf{Sp}_4(\mathbb{Z})) = \mathbb{C}[E_4, E_6, \Delta_{10}, \Delta_{12}],$$

$$\mathrm{wt}(E_4, E_6, \Delta_{10}, \Delta_{12}) = (4, 6, 10, 12).$$

$$\Delta_{10} = \Delta_5^2, \qquad \mathrm{wt}(\Delta_5) = 5, \qquad \Delta_5|_5 g = \nu_{\Delta_5}(g)\Delta_5.$$

$$\Delta_5 = \prod_{\substack{(a,b)\in(\mathbb{Z}/2\mathbb{Z})^2 \\ {}^t ab \equiv 0 \mod 2}} \nu_{a,b},$$

$$\nu_{ab}(z) = \sum_{l\in\mathbb{Z}^2} \exp\bigl(\pi i(z[l + \tfrac{1}{2}a] + {}^t bl)\bigr), \qquad z[x] = {}^t xzx.$$

For $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathbf{Sp}_4(\mathbb{Z})$, Maass's character [25]

$$\nu_{\Delta_5} : \mathbf{Sp}_4(\mathbb{Z}) \to \mathbb{C}^\times, \qquad |\nu_{\Delta_5}| = 1,$$

is determined on the standard generators by

$$\nu_{\Delta_5}\begin{pmatrix} 0 & \mathbf{I}_2 \\ -\mathbf{I}_2 & 0 \end{pmatrix} = 1,$$

$$\nu_{\Delta_5}\begin{pmatrix} \mathbf{I}_2 & B \\ 0 & \mathbf{I}_2 \end{pmatrix} = (-1)^{b_1+b_2+b_3},$$

$$\nu_{\Delta_5}\begin{pmatrix} {}^t A^{-1} & 0 \\ 0 & A \end{pmatrix} = (-1)^{(1+a_1+a_4)(1+a_2+a_3)+a_1a_4}$$

with $A = \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \in \mathbf{GL}_2(\mathbb{Z})$ and $B = \begin{pmatrix} b_1 & b_2 \\ b_2 & b_3 \end{pmatrix} \in M_{2\times 2}(\mathbb{Z})$.

Δ_5 is a weight-5 form with character ν_{Δ_5}, and $\Delta_5^2 = \Delta_{10}$ is scalar.

For

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix},$$

$$\Delta_5(Z) = \sum_{\substack{n,l,m\equiv 1 \mod 2 \\ 4nm-l^2>0 \\ n,m>0}} c_\Delta(n,l,m)\exp\bigl(\pi i(nz_1 + lz_2 + mz_3)\bigr),$$

with

$$c_\Delta(1,1,1) = 64, \qquad c_\Delta(n,l,m) \in 64\mathbb{Z}.$$

Proposition 3.1 *(Isotropic cusp coefficient).*

$$1 + \frac{1}{64}\sum_{t\geq 1} c_\Delta(1+2t,1,1)q^t = \prod_{k\geq 1}(1-q^k)^9.$$

Proof.

$$\Delta_5(Z) = \sum_{\substack{m>0 \\ m\equiv 1 \bmod 2}} \phi_{5,m}(z_1, z_2)\exp(\pi i m z_3).$$

$$\phi_{5,1} \in J_{5,\frac{1}{2}}(\nu_{\Delta_5}).$$

$$\nu_{11}(z_1, z_2) = \sum_{n\in\mathbb{Z}} (-1)^n \exp\left(\frac{\pi i}{4}(2n+1)^2 z_1 + \pi i(2n+1)z_2\right).$$

$$\nu_{11} = -q^{1/8} r^{-1/2} \prod_{n\geq 1}(1-q^{n-1}r)(1-q^n r^{-1})(1-q^n),$$

$$q = \exp(2\pi i z_1), \qquad r = \exp(2\pi i z_2).$$

$$\psi_{5,\frac{1}{2}} = \eta(z_1)^9 \nu_{11}(z_1, z_2), \qquad \eta(z_1) = q^{1/24}\prod_{n\geq 1}(1-q^n).$$

$$\frac{1}{64}\phi_{5,1}(z_1, z_2) = \psi_{5,\frac{1}{2}}(z_1, z_2) = -q^{1/2} r^{-1/2} \prod_{n\geq 1}(1-q^{n-1}r)(1-q^n r^{-1})(1-q^n)^{10}.$$

$$1 + \frac{1}{64}\sum_{t\geq 1} c_\Delta(1+2t, 1, 1) q^t = \prod_{k\geq 1}(1-q^k)^9.$$

□

Definition 3.2 (Massless divisor and determinant sign).

$$\mathcal{H}_{\mathrm{diag}} = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mid a, b \in \mathbb{H}_1 \right\}.$$

Gritsenko–Cléry [18], $N = 1$:

$$\operatorname{supp}\operatorname{div}(\Delta_5) = \mathbf{Sp}_4(\mathbb{Z}) \cdot \mathcal{H}_{\mathrm{diag}}, \qquad \operatorname{mult}_{\mathcal{H}_{\mathrm{diag}}}(\Delta_5) = 1.$$

$$\mathcal{M}_{\mathrm{massless}} := \operatorname{supp}\operatorname{div}(\Delta_5), \qquad \epsilon_{\det} := \nu_{\Delta_5}.$$

$$\Delta_5|_5 g = \nu_{\Delta_5}(g)\Delta_5, \qquad \Delta_5(s_{\delta_i} Z) = -\Delta_5(Z).$$

PROPOSITION 3.3 *(Igusa identification of the determinant).*

$$\mathcal{D}_X(Z) = \Delta_5(Z).$$

Proof. Borcherds' product theorem for $\phi_{0,1}$:

$$64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

$$\mathrm{wt} = f(0,0)/2 = 5, \qquad c_\Delta(1,1,1) = 64.$$

Gritsenko–Nikulin [19]:

$$64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} = \Delta_5(Z).$$

$$\mathcal{D}_X(Z) = \Delta_5(Z).$$

□

Remark 3.4 (Normalization of the square root). The constants in the square-root product are fixed by the following chain:

Object	Fixed value
Fourier variables	$q = e^{2\pi i z_1},\ r = e^{2\pi i z_2},\ s = e^{2\pi i z_3}$
Borcherds input	$Z_{\mathrm{K3}} = 2\phi_{0,1}$
Weight	$\mathrm{wt}\ \mathrm{Borch}(\phi_{0,1}) = f(0,0)/2 = 5$
Igusa leading coefficient	$[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$
Monic product	$D_5 = 64^{-1}\Delta_5 = q^{1/2}r^{1/2}s^{1/2}\prod(1-q^n r^l s^m)^{f(nm,l)}$
Determinant normalization	$\mathcal{D}_X = \Delta_5, \quad v_0 = 64q^{1/2}r^{1/2}s^{1/2}$
Virtual inverse	$\mathcal{Z}^{\mathrm{vir}}_{\mathrm{BPS}} = 64^{-1}q^{-1/2}r^{-1/2}s^{-1/2}\,\mathrm{sch}\,\mathcal{F}(\mathcal{V})$
BKM denominator	$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 64^{-1}\Delta_5(2Z)$
Scalar square	$Z^X_{\mathrm{OP}} = -4096\Delta_5^{-2}$; in general $Z^X_{\square} = C_{\square}\Delta_5^{-2}$, hence $(\mathcal{Z}^{\mathrm{vir}}_{\mathrm{BPS}})^2 = Z^X_{\square}/C_{\square}$

Proposition 3.5 *(Universal Borcherds–Gritsenko weight identity at $N = 1$).* Let $\phi_{0,1} \in J^{\mathrm{weak}}_{0,1}$ be the weight-zero index-one weak Jacobi form with Fourier expansion $\phi_{0,1}(z_1, z_2) = \sum_{n,l} f(n,l)\, q^n r^l$, and let $\Phi_N := \mathrm{Borch}(\phi^{(N)}_{0,1})$ denote the Gritsenko–Nikulin singular-theta lift of the N-th Hecke image. At $N = 1$ the lift coincides with Δ_5 and the Borcherds weight reads

$$\mathrm{wt}(\Delta_5) = \frac{f(0,0)}{2} = 5 = \frac{c_1(0)}{2},$$

with $c_1(0) := f(0,0) = 10$. The identity $\mathrm{wt}(\Phi_N) = c_N(0)/2$ is the universal form: the $N = 1$ row pins $(\mathrm{wt}, c_N(0)) = (5, 10)$, and the covering group attached to the reflection datum of Δ_5 is the trivial integer-weight cover over $\mathbf{Sp}_4(\mathbb{Z})$.

Proof. The Borcherds singular-theta regularisation of $\phi_{0,1}$ produces a meromorphic Siegel modular form of weight $f(0,0)/2$ on the paramodular group $\Gamma^{(N)}_{\mathrm{para}}$; at $N = 1$ this group is $\mathbf{Sp}_4(\mathbb{Z})$ by Gritsenko–Nikulin [19]. The input Jacobi form $\phi_{0,1}$ has $f(0,0) = 10$ (Eichler–Zagier, elliptic genus of K3), hence the output weight is 5. Identifying $\mathrm{Borch}(\phi_{0,1})$ with Δ_5 via the Igusa identification ($\mathcal{D}_X = \Delta_5$) fixes $\mathrm{wt}(\Delta_5) = 5$. Setting $c_1(0) := f(0,0)$ exhibits the row ($N = 1$, $\mathrm{wt} = 5$, $c_1(0) = 10$) of the Gritsenko–Cléry spread [18]; the rows at $N \in \{2, 3, 4, 6\}$ and the half/quarter-integer continuations are anchored here but proved in [18]. Integrality $\mathrm{wt}(\Delta_5) \in \mathbb{Z}$ pins the automorphy weight-multiplier to the trivial line bundle over $\mathbf{Sp}_4(\mathbb{Z})$: no passage to Mp_4 is required, and the non-trivial Maass character ν_{Δ_5} is a finite-order representation of $\mathbf{Sp}_4(\mathbb{Z})$ in its own right, orthogonal to the weight datum. □

4 The protected D-brane index

Let S be an elliptically fibered K3 surface

$$\pi : S \to \mathbb{P}^1$$

with a section s_1, and let E be an elliptic curve. Set

$$X = S \times E.$$

Let $F \in \mathrm{Pic}(S)$ be the fibre class and put

$$\beta_h = s_1 + hF, \qquad h \geq 0.$$

Type IIA D6–D2–D0 charge:

$$\gamma(h,d,n) = (1;\ (\beta_h, d);\ n).$$

Write $\Omega_X(h,d,n)$ for the protected D6–D2–D0 index attached to this charge. The microscopic lift is open:

$$\exists\, \mathcal{H}_{\gamma(h,d,n)}, \qquad \mathrm{Tr}_{\mathcal{H}_{\gamma(h,d,n)}}(-1)^F = \Omega_X(h,d,n).$$

$$\mathrm{Hilb}^n(X,(\beta_h,d)) = \{Y \subset X \mid [Y] = (\beta_h, d),\ \chi(\mathcal{O}_Y) = n\}.$$

$$E \curvearrowright X, \qquad (x, e') \mapsto (x, e + e').$$

$$\mathbf{DT}^X_{n,(\beta_h,d)} = \int_{\mathrm{Hilb}^n(X,(\beta_h,d))/E} \nu\, de,$$

$$\nu = \text{Behrend function}.$$

Behrend–Thomas: [2], [33].

5 The scalar square

$$Z^X_{\mathrm{DT}}(q,t,p) = \sum_{h=0}^{\infty}\sum_{d=0}^{\infty}\sum_{n\in\mathbb{Z}} \mathbf{DT}^X_{n,(\beta_h,d)} q^{d-1} t^{\frac{1}{2}\langle \beta_h,\beta_h\rangle}(-p)^n.$$

PROPOSITION 5.1 *(Oberdieck–Pixton normalization).* Let the Oberdieck–Pixton variables be

$$p_{\mathrm{OP}} = e^{iu}, \qquad q_{\mathrm{OP}}, \qquad \widetilde{q}_{\mathrm{OP}}.$$

The variable map is

$$q = q_{\mathrm{OP}}, \qquad r = p_{\mathrm{OP}}, \qquad s = \widetilde{q}_{\mathrm{OP}}.$$

For the displayed DT variables,

$$q_{\mathrm{DT}} = \widetilde{q}_{\mathrm{OP}} = s, \qquad t_{\mathrm{DT}} = q_{\mathrm{OP}} = q, \qquad p_{\mathrm{DT}} = p_{\mathrm{OP}} = r.$$

The product denoted χ_{10} by Oberdieck–Pixton is

$$\chi_{10}^{\mathrm{OP}} = D_5^2 = \left(64^{-1}\Delta_5\right)^2 = 4096^{-1}\Delta_5^2.$$

Consequently their primitive reduced partition function is

$$Z^X_{\mathrm{OP}} = -\left(\chi_{10}^{\mathrm{OP}}\right)^{-1} = -4096\,\Delta_5^{-2}.$$

Proof. Oberdieck–Pixton use the primitive expansion [30, Theorem 1]

$$Z^X_{\mathrm{OP}} = \sum_{g,h,d} N_{g,h,d} u^{2g-2} q_{\mathrm{OP}}^{h-1} \widetilde{q}_{\mathrm{OP}}^{d-1}$$

and the stable-pairs substitution, in the coordinate convention of Oberdieck–Pandharipande [29, Section 2],

$$y = -p_{\mathrm{OP}}, \qquad p_{\mathrm{OP}} = e^{iu}.$$

There

$$\Omega = \begin{pmatrix} \tau & z \\ z & \widetilde{\tau} \end{pmatrix}, \qquad p_{\mathrm{OP}} = e^{2\pi i z} = e^{iu}, \qquad q_{\mathrm{OP}} = e^{2\pi i \tau}, \qquad \widetilde{q}_{\mathrm{OP}} = e^{2\pi i \widetilde{\tau}}.$$

The displayed DT convention uses

$$q_{\mathrm{DT}}^{d-1} t_{\mathrm{DT}}^{h-1} (-p_{\mathrm{DT}})^n,$$

and Bryan records that the DT convention interchanges the q and $\widetilde{q}$ variables relative to Oberdieck–Pandharipande [6, Definition 2.1 and Conjecture 2.3]. This gives the stated variable map.

Oberdieck–Pixton define the Igusa product by [30, (3)]

$$\chi_{10}^{\mathrm{OP}}(p, q, \widetilde{q}) = pq\widetilde{q} \prod (1 - p^k q^h \widetilde{q}^d)^{c(4hd-k^2)}$$

in the chamber $h > 0$ or $d > 0$, together with $h = d = 0,\ k < 0$. The exponents are the K3 elliptic-genus coefficients. Since

$$Z_{\mathrm{K3}} = 2\phi_{0,1}, \qquad \phi_{0,1} = \sum f(n, l) q^n r^l,$$

those exponents are $2f(nm, l)$. Therefore the Oberdieck–Pixton monic product is exactly the square of

$$D_5 = 64^{-1}\Delta_5 = q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

Their Theorem 1, equivalently the Laurent expansion in [30, (56)], gives $Z_{\mathrm{OP}}^X = -1/\chi_{10}^{\mathrm{OP}}$. □

Theorem 5.2 *(Oberdieck–Pixton/Oberdieck reduced DT square).* Let $\beta_h \in H_2(S, \mathbb{Z})$ be primitive and non-zero, with

$$\langle \beta_h, \beta_h \rangle = 2h - 2.$$

Under the variable map

$$p = P = e^{iu}, \qquad t = Q, \qquad q = T, \qquad (q_{\mathrm{Bch}}, r_{\mathrm{Bch}}, s_{\mathrm{Bch}}) = (Q, P, T),$$

one has

$$Z_{\mathrm{DT}}^X(T, Q, P) = Z_{\mathrm{OP}}^X(P, Q, T) = -4096\, \Delta_5(Z)^{-2}.$$

Equivalently,

$$C_{\mathrm{DT}} = C_{\mathrm{OP}} = -4096.$$

Proof. Oberdieck's reduced DT/PT theorem for $S \times E$, [28, Theorem 3(i)], gives, for every non-zero $\beta \in H_2(S, \mathbb{Z})$,

$$\sum_n \mathbf{DT}_{n,(\beta,d)}^X x^n = \sum_n \mathsf{P}_{n,(\beta,d)}^X x^n.$$

Set $x = -P$. The degree-zero factor is

$$M(-P)^{\chi(S\times E)} = M(-P)^0 = 1.$$

For primitive β_h, the reduced GW/PT comparison used in Oberdieck–Pixton [30, Section 7.1, equation (GWPTcorr)], together with Oberdieck–Pandharipande [29, Proposition 5], gives

$$\sum_n \mathsf{P}_{n,(\beta_h,d)}^X (-P)^n = \sum_g \mathsf{N}_{g,h,d}^X u^{2g-2}, \qquad P = e^{iu}.$$

Multiplying by $Q^{h-1}T^{d-1}$ and summing over h, d gives

$$Z^X_{\mathrm{DT}}(T,Q,P) = Z^X_{\mathrm{OP}}(P,Q,T).$$

Oberdieck–Pixton [30, Theorem 1] and Proposition 5.1 give

$$Z^X_{\mathrm{OP}}(P,Q,T) = -4096\,\Delta_5(Z)^{-2}.$$

□

Conjecture 5.3 *(Protected D-brane index).* For the D6–D2–D0 charge

$$\gamma(h,d,n) = (1;\ (\beta_h, d);\ n),$$

the physical protected index is represented, in the conjectural enumerative identification, by the Behrend-weighted Hilbert-scheme invariant:

$$\Omega_X(h,d,n) = \mathrm{Tr}_{\mathcal{H}_{\gamma(h,d,n)}}(-1)^F \overset{\text{phys}}{=} \mathbf{DT}^X_{n,(\beta_h,d)}.$$

This conjectural identification is separate from the enumerative scalar identity in Theorem 5.2.

Definition 5.4 *(Scalar-square branch).* The datum of a scalar-square branch is a source

$$\mathfrak{s} \in \{\mathrm{OP}, \mathrm{DT}\}$$

and a Laurent expansion satisfying

$$Z^X_{\mathfrak{s}}(Z) = \frac{C_{\mathfrak{s}}}{\Delta_5(Z)^2}, \qquad C_{\mathfrak{s}} \in \mathbb{C}^\times.$$

The unconditional branch is

$$Z^X_{\square,\mathrm{uncond}} := Z^X_{\mathrm{OP}}, \qquad C_{\square,\mathrm{uncond}} := -4096.$$

The Hilbert-scheme DT branch is

$$Z^X_{\square,\mathrm{DT}} := Z^X_{\mathrm{DT}}, \qquad C_{\square,\mathrm{DT}} := -4096.$$

If the source is not displayed, $Z^X_{\square}$ denotes an arbitrary scalar-square branch.

THEOREM 5.5 *(Square-root consequence).* For $\mathfrak{s} \in \{\mathrm{OP}, \mathrm{DT}\}$, with variables normalised as in Theorem 5.2,

$$Z^X_{\mathfrak{s}}(Z) = -4096\,\Delta_5(Z)^{-2}.$$

Thus

$$\mathcal{D}_X(Z)^2 = \frac{-4096}{Z^X_{\mathfrak{s}}(Z)}, \qquad \mathcal{D}_X(Z) = \Delta_5(Z).$$

$$\mathcal{Z}^{\mathrm{vir}}_{\mathrm{BPS}} := \mathcal{D}_X^{-1}.$$

$$Z^X_{\mathfrak{s}} = -4096\,(\mathcal{Z}^{\mathrm{vir}}_{\mathrm{BPS}})^2,$$

$$\mathcal{Z}^{\mathrm{vir}}_{\mathrm{BPS}} = \frac{1}{64} q^{-1/2} r^{-1/2} s^{-1/2}\,\mathrm{sch}\,\mathcal{F}(\mathcal{V}).$$

Proof. Theorem 5.2 gives the first display for $\mathfrak{s} = \mathrm{OP}$ and $\mathfrak{s} = \mathrm{DT}$. Hence

$$-4096/Z_{\mathfrak{s}}^X = \Delta_5^2.$$

The chosen square root is the Borcherds determinant already normalized as

$$\mathcal{D}_X = \Delta_5,$$

not an arbitrary analytic branch. Inverting gives

$$Z_{\mathfrak{s}}^X = -4096(\mathcal{D}_X^{-1})^2.$$

□

Object	Equation	Claim strength
Primitive reduced curve counting	$Z_{\mathrm{OP}}^X = -4096\Delta_5^{-2}$	proved by Oberdieck–Pixton for primitive K3 classes, after Proposition 5.1
Hilbert-scheme DT series	$Z_{\mathrm{DT}}^X = \sum \mathbf{DT}^X_{n,(\beta_h,d)} T^{d-1}Q^{h-1}(-P)^n$	definition; equality with $-4096\Delta_5^{-2}$ follows from Oberdieck reduced DT/PT and Theorem 5.2
Protected D-brane index	$\Omega_X(h,d,n) \overset{\text{phys}}{=} \mathbf{DT}^X_{n,(\beta_h,d)}$	conjectural physics identification
Scalar-square branch	$Z_{\square}^X = C_{\square}\Delta_5^{-2}$	unconditional with $C_{\square} = -4096$ for Z_{OP}^X and Z_{DT}^X

6 The exterior-square model of the duality group

$$\Lambda^4 = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}e_3 \oplus \mathbb{Z}e_4.$$

$$g \in \mathbf{SL}_4(\mathbb{Z}) : \qquad \wedge^2 g : \wedge^2\Lambda^4 \longrightarrow \wedge^2\Lambda^4.$$

$$u \wedge v = (u,v)e_1 \wedge e_2 \wedge e_3 \wedge e_4.$$

$$\big(\wedge^2\Lambda^4, (\ ,\ )\big) \simeq \mathrm{II}_{3,3}.$$

Fix

$$q_0 = e_1 \wedge e_3 + e_2 \wedge e_4.$$

For $x, y \in \Lambda^4$, define

$$-x \wedge y \wedge q_0 = B_{q_0}(x,y)e_1 \wedge e_2 \wedge e_3 \wedge e_4.$$

$$\{g \in \mathbf{SL}(\Lambda^4) \mid (\wedge^2 g)q_0 = q_0\} \simeq \mathbf{Sp}_4(\mathbb{Z}).$$

Set

$$\Lambda^{3,2} = q_0^{\perp} \subset \wedge^2\Lambda^4.$$

$$f_1 = e_1 \wedge e_2, \quad f_2 = e_2 \wedge e_3, \quad f_3 = e_1 \wedge e_3 - e_2 \wedge e_4, \quad f_{-2} = e_4 \wedge e_1, \quad f_{-1} = e_4 \wedge e_3.$$

$$(f_1, f_{-1}) = (f_2, f_{-2}) = -1, \qquad (f_3, f_3) = 2.$$

$$\Lambda^{3,2} \simeq \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2], \qquad \Lambda^{(1,1)} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

$$\mathbb{H}^{\mathbf{IV}} = \{Z \in \mathbb{P}(\Lambda^{3,2} \otimes \mathbb{C}) \mid (Z, Z) = 0,\ (Z, \overline{Z}) < 0\}.$$
$$\mathbb{H}^{\mathbf{IV}}_{+} = \{{}^t(z_2^2 - z_1 z_3,\ z_3,\ z_2,\ z_1,\ 1) z_0 \in \mathbb{H}^{\mathbf{IV}} \mid \mathrm{Im} z_1 > 0\}.$$

Writing $y_i = \mathrm{Im} z_i$,

$$(Z, \overline{Z}) < 0 \quad \Longleftrightarrow \quad y_1 y_3 - y_2^2 > 0.$$
$$\mathbb{H}^{\mathbf{IV}}_{+} \simeq \left\{ \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix} \in \mathbb{H}_2 \right\}.$$

Let $\mathbf{O}_{\mathbb{R}}(\Lambda^{3,2})_+$ be the index-two subgroup preserving $\mathbb{H}^{\mathbf{IV}}_{+}$. Its kernel on $\mathbb{H}^{\mathbf{IV}}_{+}$ is $\{\pm \mathbf{I}_5\}$, and

$$\mathbf{O}_{\mathbb{R}}(\Lambda^{3,2})_+ = \{\pm \mathbf{I}_5\} \mathbf{SO}_{\mathbb{R}}(\Lambda^{3,2})_+,$$

where $\mathbf{SO}_{\mathbb{R}}(\Lambda^{3,2})_+$ has real spin-norm 1. Put

$$\mathbf{O}(\Lambda^{3,2})_+ = \mathbf{O}_{\mathbb{R}}(\Lambda^{3,2})_+ \cap \mathbf{O}(\Lambda^{3,2})$$

and

$$\mathbf{SO}(\Lambda^{3,2})_+ = \mathbf{SO}_{\mathbb{R}}(\Lambda^{3,2})_+ \cap \mathbf{O}(\Lambda^{3,2}).$$

$$M = \begin{pmatrix} m_1 & m_2 \\ m_2 & m_3 \end{pmatrix},$$

$$\wedge^2(g_0) = \wedge^2\left(\begin{pmatrix} 0 & \mathbf{I}_2 \\ -\mathbf{I}_2 & 0 \end{pmatrix}\right) = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\wedge^2(g_M) = \wedge^2\left(\begin{pmatrix} \mathbf{I}_2 & M \\ 0 & \mathbf{I}_2 \end{pmatrix}\right) = \begin{pmatrix} 1 & -m_1 & 2m_2 & -m_3 & m_2^2 - m_1 m_3 \\ 0 & 1 & 0 & 0 & m_3 \\ 0 & 0 & 1 & 0 & m_2 \\ 0 & 0 & 0 & 1 & m_1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

$$\wedge^2(g_A) = \wedge^2\left(\begin{pmatrix} {}^tA^{-1} & 0 \\ 0 & A \end{pmatrix}\right) = \det(A) \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & a_1^2 & -2a_1a_2 & a_2^2 & 0 \\ 0 & -a_1a_3 & a_1a_4 + a_2a_3 & -a_2a_4 & 0 \\ 0 & a_3^2 & -2a_3a_4 & a_4^2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

LEMMA 6.1. The exterior-square representation is

$$\wedge^2 : \mathbf{Sp}_4(\mathbb{Z})/\{\pm \mathbf{I}_4\} \to \mathbf{SO}_+(\Lambda^{3,2}) \simeq \mathbf{O}(\Lambda^{3,2})_+/\{\pm \mathbf{I}_5\}.$$

For every $g \in \mathbf{Sp}_4(\mathbb{Z})$,

$$\begin{tikzcd} \mathbb{H}_2 \arrow[r] \arrow[d] & \mathbb{H}_2 \arrow[d] \\ \mathbb{H}_+^{\mathbf{IV}} \arrow[r, "g\wedge g"] & \mathbb{H}_+^{\mathbf{IV}} \end{tikzcd}$$

Proof. The Pfaffian pairing is preserved by $\wedge^2\mathbf{SL}_4$. The subgroup fixing

$$q_0 = e_1 \wedge e_3 + e_2 \wedge e_4$$

is the group preserving the alternating form B_{q_0}, namely $\mathbf{Sp}_4(\mathbb{Z})$. Its action preserves $q_0^\perp = \Lambda^{3,2}$ and the displayed component of the type-**IV** domain. The kernel on $q_0^\perp$ is $\{\pm\mathbf{I}_4\}$. The exceptional isomorphism $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}_{\mathbb{R}}(3,2)_+$ identifies the image with $\mathbf{SO}_+(\Lambda^{3,2})$. The displayed matrices are the integral generators on both sides. Substitution of

$${}^t(z_2^2 - z_1z_3, z_3, z_2, z_1, 1)z_0$$

into the two actions produces the commutative square. □

Definition 6.2 *(Automorphic duality datum).*

$$G_{\mathrm{aut}} = \mathbf{PSp}_4(\mathbb{Z}) \simeq \mathbf{SO}_+(\Lambda^{3,2}).$$

The protected determinant is not an invariant scalar function on $\mathbb{H}_2$. It is a section of the automorphic line

$$\mathcal{L}^5 \otimes \nu_{\Delta_5}$$

with transformation law

$$\Delta_5(g \cdot Z) = \nu_{\Delta_5}(g)\det(CZ + D)^5\Delta_5(Z), \qquad g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathbf{Sp}_4(\mathbb{Z}).$$

The scalar square is a section of $\mathcal{L}^{10}$:

$$\Delta_5^2(g \cdot Z) = \det(CZ + D)^{10}\Delta_5^2(Z).$$

For every scalar-square branch,

$$Z_\square^X(g \cdot Z) = \det(CZ + D)^{-10}Z_\square^X(Z).$$

7 The Weyl chamber and the Weyl vector

Change polarization on the same Gram-index lattice:

$$\Gamma_{\mathrm{ind}} \xrightarrow{\alpha} \Lambda_{II}^{2,1}, \qquad \alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}.$$

The cusp polarization gives the Fock determinant. The denominator polarization gives the Borcherds chamber.

Fix the primitive hyperbolic sublattice

$$\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2] \simeq \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2} \subset \Lambda^{3,2}$$

Extend $\phi \in \mathbf{O}(\Lambda^{2,1})$ by the identity on $(\Lambda^{2,1})^{\perp}$:

$$\mathbf{O}(\Lambda^{2,1}) \hookrightarrow \mathbf{O}(\Lambda^{3,2}).$$

$$\Delta_5 \in M_5(\mathbf{O}(\Lambda^{2,1}), \nu_{\Delta_5}).$$

For a primitive element $\alpha \in \Lambda^{2,1}$ satisfying $(\alpha, \alpha) > 0$ and

$$(\alpha, \alpha) \mid 2(\Lambda^{2,1}, \alpha),$$

Kac reflection [23]:

$$s_\alpha : x \mapsto x - 2\frac{(x, \alpha)}{(\alpha, \alpha)}\alpha.$$

$$s_\alpha(\alpha) = -\alpha, \qquad s_\alpha|_{\alpha^\perp} = \mathbf{I}.$$

For $(\alpha, \alpha) = 2$:

$$s_\alpha : x \mapsto x - (x, \alpha)\alpha.$$

LEMMA 7.1. Let $(\alpha, \alpha) = 2$. For

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3,$$

$$\Delta_5(s_\alpha Z) = -\Delta_5(Z)$$

when $\alpha \in \{\delta_1, \delta_2, \delta_3\}$. For

$$\alpha \in \{f_{-2} - f_2,\ f_2 - f_3,\ f_2 + f_3\},$$

$$\Delta_5(s_\alpha Z) = \Delta_5(Z)$$

Proof. Let

$$\overline{U} = \wedge^2 \begin{pmatrix} {}^tU^{-1} & 0 \\ 0 & U \end{pmatrix}, \qquad U \in \mathbf{GL}_2(\mathbb{Z}).$$

Required reflections:

$$s_{f_{-2}-f_2} = -\overline{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}, \qquad s_{f_3} = -\overline{\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}, \qquad s_{f_2-f_3} = -\overline{\begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}}.$$

Maass's character has value -1 on s_{δ_i} and $+1$ on the three other reflection classes. □

The time-like cone is

$$C(\Lambda^{2,1}) = \{x \in \Lambda^{2,1} \otimes \mathbb{R} | (x, x) < 0\}.$$

Let $C(\Lambda^{2,1})_+$ be the component for which

$$\Omega(C(\Lambda^{2,1})_+) := \Lambda^{2,1} \otimes \mathbb{R} + iC(\Lambda^{2,1})_+ \subset \mathbb{H}_+^{\mathbf{IV}}.$$

For

$$z = z_3 f_2 + z_2 f_3 + z_1 f_{-2} \in \Omega(C(\Lambda^{2,1})_+),$$

the type **IV** point is

$$Z = {}^t(z_2^2 - z_1 z_3,\ z_3,\ z_2,\ z_1,\ 1)z_0.$$

Let $\mathbf{O}(\Lambda^{2,1})_+$ be the index-two subgroup fixing $C(\Lambda^{2,1})_+$. It acts discretely on

$$C(\Lambda^{2,1})_+/\mathbb{R}_{>0}.$$

For $(\alpha,\alpha) > 0$,

$$\mathbb{H}_\alpha = \{\mathbb{R}_{>0}x \in C(\Lambda^{2,1})_+/\mathbb{R}_{>0} \mid (x,\alpha) = 0\},$$

and

$$\mathbb{H}_{\alpha,+} = \{\mathbb{R}_{>0}x \in C(\Lambda^{2,1})_+/\mathbb{R}_{>0} \mid (x,\alpha) \le 0\}.$$

Let $W(\Lambda^{2,1})$ be the subgroup generated by these positive-norm reflections:

$$W(\Lambda^{2,1}) \subset \mathbf{O}(\Lambda^{2,1})_+.$$

Nikulin's theorem for $\Lambda^{(1,1)} \oplus [2]$ [27]:

$$\mathbf{O}(\Lambda^{2,1})_+ = W^{(2)}(\Lambda^{2,1}),$$

where $W^{(2)}$ is generated by reflections in square-2 vectors. For $k > 0$ put

$$\Delta^{(k)}(\Lambda^{2,1}) = \{\delta \in \Lambda^{2,1} \mid \delta \text{ primitive},\ (\delta,\delta) = k,\ s_\delta \in \mathbf{O}(\Lambda^{2,1})\}.$$

Each $\delta \in \Delta^{(2)}(\Lambda^{2,1})$ has one of two divisibilities:

- Type I: $(\delta, \Lambda^{2,1}) = \mathbb{Z}$
- Type II: $(\delta, \Lambda^{2,1}) = 2\mathbb{Z}$

The type sublattices are

$$\Lambda_I^{2,1} = \{mf_2 + lf_3 + nf_{-2} \in \Lambda^{2,1} \mid m + l + n \equiv 0 \pmod 2\},$$

$$\Lambda_{II}^{2,1} = \{mf_2 + lf_3 + nf_{-2} \in \Lambda^{2,1} \mid m \equiv n \equiv 0 \pmod 2\}.$$

Type I roots lie in $\Lambda_I^{2,1}$; type II roots lie in $\Lambda_{II}^{2,1}$. The reflection subgroups are normal, with indices

$$[W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda_I^{2,1})] = 2, \qquad [W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda_{II}^{2,1})] = 6.$$

The chambers are

$$\mathcal{P}' = \bigcap_{\delta \in \mathcal{P}'_{prim}} \mathbb{H}_{\delta,+}, \qquad \mathcal{P}' \in \{\mathcal{P}, \mathcal{P}_I, \mathcal{P}_{II}\},$$

with walls

$$\mathcal{P}_{prim} = \{f_2 - f_3,\ f_{-2} - f_2,\ f_3\},$$
$$\mathcal{P}_{I,prim} = \{f_2 - f_3,\ f_{-2} - f_2,\ f_2 + f_3\},$$
$$\mathcal{P}_{II,prim} = \{\delta_1, \delta_2, \delta_3\}.$$

For

$$\mathrm{Aut}(\mathcal{P}') = \{g \in \mathbf{O}(\Lambda^{2,1})_+ \mid g\mathcal{P}' = \mathcal{P}'\},$$
$$\mathrm{Aut}(\mathcal{P}) = 1, \qquad \mathrm{Aut}(\mathcal{P}_I) = \langle s_{f_3}\rangle, \qquad \mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3,$$

with $\mathrm{Aut}(\mathcal{P}_{II})$ generated by $s_{f_2-f_3}$ and $s_{f_{-2}-f_2}$. The chamber decomposition is

$$\mathbf{O}(\Lambda^{2,1})_+ \simeq W^{(2)}(\Lambda^{2,1}) \simeq W^{(2)}(\Lambda_I^{2,1}) \rtimes \mathrm{Aut}(\mathcal{P}_I) \simeq W^{(2)}(\Lambda_{II}^{2,1}) \rtimes \mathrm{Aut}(\mathcal{P}_{II}).$$

LEMMA 7.2 *(Multiplier restriction).* • for $w \in W^{(2)}(\Lambda_I^{2,1})$ and $a \in \mathrm{Aut}(\mathcal{P}_I)$,

$$\Delta_5(w \cdot aZ) = \det(a)\Delta_5(Z)$$

• for $w \in W^{(2)}(\Lambda_{II}^{2,1})$ and $a \in \mathrm{Aut}(\mathcal{P}_{II})$,

$$\Delta_5(w \cdot aZ) = \det(w)\Delta_5(Z)$$

Proof. Displayed Maass multiplier:

$$\nu_{\Delta_5}(s_{\delta_i}) = -1, \qquad \nu_{\Delta_5}(s_{f_{-2}-f_2}) = \nu_{\Delta_5}(s_{f_2-f_3}) = \nu_{\Delta_5}(s_{f_2+f_3}) = 1.$$

Semidirect products:

$$W^{(2)}(\Lambda_I^{2,1}) \rtimes \mathrm{Aut}(\mathcal{P}_I), \qquad W^{(2)}(\Lambda_{II}^{2,1}) \rtimes \mathrm{Aut}(\mathcal{P}_{II}).$$

□

Set

$$\Delta(\Lambda_{II}^{2,1})_+ = \mathbb{R}_{\geq 0}\delta_1 + \mathbb{R}_{\geq 0}\delta_2 + \mathbb{R}_{\geq 0}\delta_3$$

and

$$\Delta(\Lambda_{II}^{2,1})_+^* = \{x \in \Lambda^{2,1} \otimes \mathbb{R} | (x, \delta_i) \leq 0\}.$$

For the finite-volume chamber $\mathcal{P}_{II}$, the dual cone obeys

$$\Delta(\Lambda_{II}^{2,1})_+^* \subset \overline{C(\Lambda^{2,1})_+} \subset \Delta(\Lambda_{II}^{2,1})_+.$$

The Weyl vector is determined by

$$(\rho, \delta_i) = -\frac{(\delta_i, \delta_i)}{2} = -1, \qquad i = 1, 2, 3.$$

$$(\delta_i, \delta_j) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

$$\rho = \frac{1}{2}\delta_1 + \frac{1}{2}\delta_2 + \frac{1}{2}\delta_3 = f_2 - \frac{1}{2}f_3 + f_{-2}.$$

Also

$$\rho \in C(\Lambda^{2,1})_+ = C(\Lambda_{II}^{2,1})_+.$$

Fix

$$z = z_1 f_{-2} + z_2 f_3 + z_3 f_2 \in \Lambda^{2,1} \otimes \mathbb{R} + iC(\Lambda^{2,1})_+ = \Lambda_{II}^{2,1} \otimes \mathbb{R} + iC(\Lambda_{II}^{2,1})_+.$$

The dual lattice is

$$(\Lambda_{II}^{2,1})^* = \mathbb{Z}\frac{1}{2}f_2 + \mathbb{Z}\frac{1}{2}f_3 + \mathbb{Z}\frac{1}{2}f_{-2} = \frac{1}{2}\Lambda^{2,1}.$$

Definition 7.3 (Gram-index degrees in the Weyl chamber). For $\gamma = (n, l, m) \in \Gamma_{\mathrm{ind}}$ set

$$\alpha(\gamma) = \alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2} \in \Lambda_{II}^{2,1}.$$

With

$$z = z_1 f_{-2} + z_2 f_3 + z_3 f_2,$$

the character of the Gram-index charge is

$$x^\gamma = q^n r^l s^m = \exp(-\pi i(\alpha(\gamma), z)).$$

For $\gamma' = (n', l', m')$,

$$(\alpha(\gamma), \alpha(\gamma')) = -2\langle \gamma, \gamma' \rangle_{\mathrm{ind}}, \qquad (\alpha(\gamma), \alpha(\gamma)) = -2(4nm - l^2).$$

Thus the Borcherds degree is the Lorentzian image of the Gram-index grading. The Weyl chamber is determined by the cusp polarization fixed in Section 1 together with the active Jacobi support, since charges with zero exponent do not generate walls.

$$E_0 := \rho.$$

$$q^{1/2} r^{1/2} s^{1/2} = \exp(-\pi i(\rho, z)).$$

$$\exp(-2\pi i(\alpha(\gamma), z)), \qquad \exp(-2\pi i(\rho, z)).$$

Definition 7.4 (Active support and Borcherds cone). In the standard cusp chamber, the active charge support is

$$\Gamma_{\mathrm{act}} = \{\gamma = (n, l, m) \in \Gamma_{\mathrm{eff}} \mid f(nm, l) \neq 0\}.$$

The inactive triples in $\Gamma_{\mathrm{eff}} \setminus \Gamma_{\mathrm{act}}$ are ordering data only; their product exponents are zero. The product does not prove absence of root spaces with zero signed superdimension. The proof-grade Borcherds cone is the real cone generated by the active positive Gram-index charges after the Lorentzian degree map:

$$C_{\mathrm{Borch}} := \mathbb{R}_{\geq 0}\, \alpha(\Gamma_{\mathrm{act}}) \subset \Lambda_{II}^{2,1} \otimes \mathbb{R}.$$

The real-root hyperplanes are the walls used by the denominator identity in this polarization.

Lemma 7.5 *(Active support gives the type-II chamber cone).* For $\gamma = (n, l, m) \in \Gamma_{\mathrm{act}}$,

$$\alpha(\gamma) = n\delta_1 + m\delta_2 + (n + m - l)\delta_3,$$

with

$$n \geq 0, \qquad m \geq 0, \qquad n + m - l \geq 0.$$

Consequently

$$C_{\mathrm{Borch}} = \mathbb{R}_{\geq 0}\delta_1 + \mathbb{R}_{\geq 0}\delta_2 + \mathbb{R}_{\geq 0}\delta_3.$$

Proof. The displayed decomposition is

$$n(2f_2 - f_3) + m(2f_{-2} - f_3) + (n + m - l)f_3 = 2nf_2 - lf_3 + 2mf_{-2}.$$

The effective chamber gives $n, m \geq 0$. Since $\phi_{0,1}$ is weak of index one, $f(nm, l) \neq 0$ implies $4nm - l^2 \geq -1$. If $n = 0$ or $m = 0$, this gives $l^2 \leq 1$, and the boundary cases in Γ_{eff} give $l \leq n + m$. If $n, m > 0$ and $l \geq n + m + 1$, then

$$l^2 \geq (n + m + 1)^2 = 4nm + (n - m)^2 + 2(n + m) + 1 > 4nm + 1,$$

contradicting $l^2 \leq 4nm + 1$. Hence $n + m - l \geq 0$. The reverse inclusion follows from the active charges

$$(1, 0, 0), \qquad (0, 0, 1), \qquad (0, -1, 0),$$

whose degrees are $\delta_1 + \delta_3$, $\delta_2 + \delta_3$, and δ_3, and from the active charges

$$(1, 1, 0), \qquad (0, 1, 1),$$

whose degrees are δ_1 and δ_2. □

For $n, l, m \in \mathbb{Z}$ with $n \equiv m \equiv l \equiv 1 \mod 2$, $n, m > 0$ and $4nm - l^2 > 0$, set

$$a = (n - 1)f_2 - \frac{l - 1}{2} f_3 + (m - 1)f_{-2}, \qquad m(a) = -\frac{1}{64} c_\Delta(n, l, m).$$

$$\rho + a = nf_2 - \frac{l}{2} f_3 + mf_{-2},$$

and

$$\begin{aligned} &\frac{1}{64} c_\Delta(n, l, m) \exp(\pi i(nz_1 + lz_2 + mz_3)) \\ &= \frac{1}{64} c_\Delta(n, l, m) \exp\left(-\pi i\left(nf_2 - \frac{l}{2} f_3 + mf_{-2}, z\right)\right) \\ &= -m(a) \exp(-\pi i(\rho + a, z)), \end{aligned}$$

with

$$a \in (\Lambda^{2,1})^* = \frac{1}{2} \Lambda^{2,1}_{II}.$$

The normalization $c_\Delta(1, 1, 1) = 64$ fixes $m(0) = -1$. Since $c_\Delta(n, l, m) \in 64\mathbb{Z}$, all $m(a)$ are integral. The inequality $4nm - l^2 > 0$ implies

$$\rho + a \in C(\Lambda^{2,1})_+.$$

Let $\lambda = \rho + a$. For nonzero coefficients in the chosen chamber,

$$\lambda \in (\Lambda^{2,1}_{II})^* \cap \mathbb{R}_+ \mathcal{P}_{II}, \qquad (\lambda, \delta_i) \leq 0.$$

If $(\lambda, \delta_i) = 0$, then

$$c_\lambda = -c_\lambda = 0.$$

For nonzero terms, $(\lambda, \delta_i) < 0$. Since $(\rho, \delta_i) = -1$,

$$(a, \delta_i) \leq 0, \qquad a \in \Delta(\Lambda^{2,1}_{II})^*_+.$$

$$a \in \mathbb{R}_{\geq 0} C(\Lambda^{2,1}_{II})^*_+ \cap \overline{C(\Lambda^{2,1})_+} = \mathbb{R}_{\geq 0} \mathcal{P}_{II}.$$

The congruences $m \equiv n \equiv l \equiv 1 \pmod 2$ imply

$$a \in 2(\Lambda^{2,1})^* = \Lambda^{2,1}_{II}.$$

$$a = 0 \qquad \text{or} \qquad a \in \Lambda^{2,1}_{II} \cap \mathbb{R}_{>0} \mathcal{P}_{II}.$$

Lemma 7.6.

$$\frac{1}{64}\Delta_5(Z) = \sum_{w\in W^{(2)}(\Lambda_{II}^{2,1})} \det(w)(\exp(-\pi i(w(\rho), z)) - \sum_{a\in\Lambda_{II}^{2,1}\cap\mathbb{R}_{>0}\mathcal{P}_{II}} m(a)\exp(-\pi i(w(\rho+a), z)))$$

Proof. Let c_λ be the coefficient of $e^{-\pi i(\lambda,z)}$ in $64^{-1}\Delta_5$. The leading term has

$$\lambda = \rho, \qquad c_\rho = 1.$$

The other exponents in the chosen chamber have

$$\lambda = \rho + a, \qquad a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{>0}\mathcal{P}_{II}, \qquad c_{\rho+a} = -m(a).$$

Anti-invariance:

$$c_{w\lambda} = \det(w)c_\lambda.$$

If λ lies on a simple wall, then

$$s_{\delta_i}\lambda = \lambda, \qquad c_\lambda = -c_\lambda = 0.$$

Summing the signed Weyl translates proves the formula. □

The primitive isotropic rays of $\overline{\mathcal{P}_{II}}$ are generated by

$$2f_2, \qquad 2f_{-2}, \qquad 2f_2 - 2f_3 + 2f_{-2}.$$

The group $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$ is transitive on these three vectors. For $a_0 = 2f_2$:

$$1 + \frac{1}{64}\sum_{t\geq 1} c_\Delta(1+2t, 1, 1)q^t = \prod_{k\geq 1}(1-q^k)^9$$

In denominator notation,

$$1 - \sum_{t\geq 1} m(ta_0)q^t = \prod_{t\geq 1}(1-q^t)^{\tau(ta_0)}, \qquad \tau(ta_0) = 9 \quad (t \geq 1).$$

By S_3-symmetry,

$$\tau(ta) = 9 \qquad (t \geq 1)$$

for every primitive isotropic ray

$$\mathbb{Z}_{>0}a \subset \overline{\mathcal{P}_{II}} \cap \Lambda_{II}^{2,1}.$$

On isotropic rays, the coefficients $m(ta)$ are determined by the eta-product expansion; the signed simple-root supermultiplicity is $\tau(ta)$.

The real root datum is the wall set

$$\mathcal{P}_{II,prim} = \{\delta_1, \delta_2, \delta_3\}$$

with Gram matrix

$$(\delta_i, \delta_j) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$$

Definition 7.7 (Perturbative walls).

$$\Delta^{re} = \{\delta_1, \delta_2, \delta_3\}, \qquad \mathcal{W}_{\mathrm{pert}} = \bigcup_{\delta\in\Delta^{re}} \mathbb{H}_\delta.$$

8 The denominator algebra

$$A = ((\delta_i, \delta_j))_{1\le i,j\le 3} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$$

$$A \text{ symmetric generalized Cartan}, \qquad \mathfrak{g}(A) \text{ Kac–Moody}.$$

$$\mathcal{A}_{\mathrm{den}} := \mathfrak{g}_{\Delta_5}.$$

This is the algebraic denominator square root. It is the automorphic correction of $\mathfrak{g}(A)$ constructed by Gritsenko–Nikulin [19], with root spaces graded by the Gram-degree map

$$\gamma = (n, l, m) \longmapsto \alpha(\gamma) = 2nf_2 - lf_3 + 2mf_{-2}.$$

The Borcherds correction character in the coefficient chamber is

$$C_{\mathrm{im}}(z) = 1 - \sum_{a\in\Lambda_{II}^{2,1}\cap\mathbb{R}_{>0}\mathcal{P}_{II}} m(a)e^{-2\pi i(a,z)}.$$

$$e^{-2\pi i(\rho,z)}C_{\mathrm{im}}(z).$$

Harvey–Moore charge grading [21]:

$$\mathrm{smult}(\alpha) = \dim \mathfrak{g}_{\alpha,\bar{0}} - \dim \mathfrak{g}_{\alpha,\bar{1}}.$$

Definition 8.1 *(Denominator Lie superalgebra).* For $\gamma \in \Gamma_{\mathrm{ind}}$ set

$$\mathcal{A}_{\mathrm{den},\gamma} := \begin{cases} \Lambda_{II}^{2,1} \otimes \mathbb{R}, & \gamma = 0, \\ \mathfrak{g}_{\Delta_5,\alpha(\gamma)}, & \alpha(\gamma) \in \mathcal{R}, \\ 0, & \alpha(\gamma) \notin \mathcal{R}. \end{cases}$$

The zero degree is the Cartan subalgebra and is not governed by $f(0, 0)$. The bracket is the Borcherds–Kac–Moody bracket of $\mathfrak{g}_{\Delta_5}$:

$$[\ ,\ ] : \mathcal{A}_{\mathrm{den},\gamma} \otimes \mathcal{A}_{\mathrm{den},\gamma'} \longrightarrow \mathcal{A}_{\mathrm{den},\gamma+\gamma'}.$$

THEOREM 8.2 *(Denominator algebra square root).* The denominator Lie superalgebra has the following properties.

$$\mathcal{A}_{\mathrm{den}} = \mathcal{A}_{\mathrm{den},0} \oplus \bigoplus_{\substack{\gamma\in\Gamma_{\mathrm{ind}} \\ \alpha(\gamma)\in\mathcal{R}}} \mathcal{A}_{\mathrm{den},\gamma}, \qquad [\mathcal{A}_{\mathrm{den},\gamma}, \mathcal{A}_{\mathrm{den},\gamma'}] \subset \mathcal{A}_{\mathrm{den},\gamma+\gamma'}.$$

Its visible positive signed charge multiplicities are the K3 half-index coefficients:

$$\mathrm{smult}(\alpha(n, l, m)) = f(nm, l) \qquad ((n, l, m) \in \Gamma_{\mathrm{act}}).$$

Its Weyl–Kac–Borcherds denominator is

$$\mathrm{den}(\mathcal{A}_{\mathrm{den}}) = 64^{-1}\Delta_5(2Z).$$

The theorem concerns the Borcherds denominator algebra. A microscopic BPS Hilbert space and its operator product require separate construction.

Proof. Gritsenko–Nikulin construct $\mathfrak{g}_{\Delta_5}$ as the generalized Borcherds–Kac–Moody superalgebra whose real simple roots are $\delta_1, \delta_2, \delta_3$ and whose imaginary simple-root multiplicities are read from the Borcherds correction character. A root-graded Lie superalgebra has

$$[\mathfrak{g}_{\Delta_5,\alpha}, \mathfrak{g}_{\Delta_5,\beta}] \subset \mathfrak{g}_{\Delta_5,\alpha+\beta}.$$

The degree map $\alpha(\gamma)$ is additive, hence the bracket conserves the Gram-degree grading. The denominator identity reads

$$\frac{1}{64}\Delta_5(2Z) = e^{-2\pi i(\rho,z)} \prod_{\alpha\in\mathcal{R}_+} (1 - e^{-2\pi i(\alpha,z)})^{\mathrm{smult}(\alpha)}.$$

Comparing this with the Borcherds product gives

$$\mathrm{smult}(\alpha(n,l,m)) = f(nm,l) \qquad ((n,l,m) \in \Gamma_{\mathrm{act}}).$$

□

Definition 8.3 *(Formal current envelope).* Let E be the elliptic curve in $X = S \times E$. Put

$$\mathrm{Cur}_E(\mathcal{A}_{\mathrm{den}}) := \mathcal{A}_{\mathrm{den}} \otimes \mathcal{O}_E$$

with the level-zero current bracket

$$[a \otimes f, b \otimes g] = [a,b] \otimes fg.$$

Its chiral enveloping algebra in the sense of Beilinson–Drinfeld [3] is

$$\mathsf{Den}_{\Delta_5,E} := U_E^{\mathrm{ch}}(\mathrm{Cur}_E(\mathcal{A}_{\mathrm{den}})).$$

Theorem 8.4 *(Formal current square root).* $\mathsf{Den}_{\Delta_5,E}$ is the formal Beilinson–Drinfeld current envelope on E obtained from the denominator Lie superalgebra. Its constant-mode primitive Lie superalgebra is $\mathcal{A}_{\mathrm{den}}$, and the denominator of that primitive bracket is

$$64^{-1}\Delta_5(2Z).$$

A BPS factorization algebra of the compactification requires additional microscopic input.

Proof. The current construction sends any Lie superalgebra to a Lie-* algebra on a smooth curve, and the chiral enveloping construction sends a Lie-* algebra to a chiral algebra [3]; equivalently, in vertex-algebra language, this is the universal current vertex algebra at level zero [22]. Constant sections of $\mathcal{A}_{\mathrm{den}} \otimes \mathcal{O}_E$ recover the original Lie superalgebra. The denominator statement is exactly Theorem 8.2. □

Definition 8.5 *(Sectorial Hall truncation).* Let $X = K3 \times E$. Fix a Bridgeland stability condition $\sigma = (Z_\sigma, \mathcal{P}_\sigma)$ on the compact category and a strict sector $S \subset \mathbb{C}^\times$ of angular width $< \pi$, with no active BPS ray on its boundary. Choose ϑ so that

$$h_S(\gamma) := \Re(e^{-i\vartheta} Z_\sigma(\gamma)) > 0$$

on every nonzero semistable class with central charge in S. For $R > 0$, set

$$F_{\sigma,S}(R) = \{\gamma \mid \mathfrak{M}^{ss}_{\sigma,S,\gamma} \neq \varnothing,\ 0 < h_S(\gamma) \leq R\},$$

and

$$\Gamma^{HN}_{\sigma,S}(R) = \left\{ \gamma_1 + \cdots + \gamma_k \,\middle|\, \gamma_i \in F_{\sigma,S}(R),\ \sum_i h_S(\gamma_i) \le R \right\}.$$

If an oriented critical Hall object $\mathcal{H}^{or}_{\sigma,S}$ has been constructed, put

$$\mathcal{H}^{>R}_{\sigma,S} = \prod_{h_S(\gamma)>R} \mathcal{H}^{or}_{\sigma,S,\gamma}, \qquad \mathsf{F}_R\mathcal{H}^{or}_{\sigma,S} = \mathcal{H}^{or}_{\sigma,S}/\mathcal{H}^{>R}_{\sigma,S}.$$

PROPOSITION 8.6 *(The finite object).* A nonzero additive heart cannot have finite nonzero charge support in Γ_{ind}. The active support Γ_{act} is not additive. The finite Hall object is therefore not a finite active-support heart.

Assume the support property, the HN property, finite type of the semistable stacks in bounded h_S-height, and the oriented critical Hall correspondences. Then $F_{\sigma,S}(R)$ and $\Gamma^{HN}_{\sigma,S}(R)$ are finite, $\mathcal{H}^{>R}_{\sigma,S}$ is a two-sided Hall ideal, and $\mathsf{F}_R\mathcal{H}^{or}_{\sigma,S}$ is a finite Hall quotient. The compact sectorial Hall object is the completed inverse system

$$\widehat{\mathcal{H}}^{or}_{\sigma,S} = \varprojlim_R \mathsf{F}_R\mathcal{H}^{or}_{\sigma,S}.$$

Proof. If a heart contains an object of nonzero charge γ, it contains all finite direct sums of that object, hence all charges $k\gamma$. Since Γ_{ind} is torsion-free, the charge support is infinite. Also $(0,-1,0) \in \Gamma_{\mathrm{act}}$, while $(0,-2,0) \notin \Gamma_{\mathrm{act}}$, so active support is not additive.

For semistable classes in S, the strict-sector condition gives $h_S(\gamma) \ge c|Z_\sigma(\gamma)|$ for some $c > 0$. The support property and discreteness of the numerical lattice make $F_{\sigma,S}(R)$ finite. The function h_S is positive and additive on extensions in the sector; hence HN sums of total height $\le R$ form a finite set and the subspace of height $> R$ is a two-sided Hall ideal. The inverse limit is the h_S-adic Hall completion. □

Definition 8.7 *(Compact Igusa realization datum).* A compact realization of the Igusa square root is a quintuple

$$\mathfrak{K}_X = (\mathcal{F}_X, \mathcal{H}_X, C_X, \Theta_{\mathrm{Kos}}, \Pi_{\mathrm{prim}})$$

with the following data.

(i) A charge-completed compact observable factorization algebra $\mathcal{F}_X$ on $\mathrm{Ran}(E)$, finite-first with respect to the sectorial HN quotients of Proposition 8.6.

(ii) An oriented compact Hall factorization object $\mathcal{H}_X$ with extension correspondences

$$\mathfrak{M}^\sigma_\gamma \times \mathfrak{M}^\sigma_\eta \xleftarrow{p_{\gamma,\eta}} \mathfrak{E}^\sigma_{\gamma,\eta} \xrightarrow{q_{\gamma,\eta}} \mathfrak{M}^\sigma_{\gamma+\eta},$$

vanishing-cycle coefficients, Thom–Sebastiani orientation transport, protected integration, and orientation character $\epsilon_o = \nu_{\Delta_S}$.

(iii) A conilpotent chiral coalgebra C_X and a charge-preserving quasi-isomorphism

$$C_X \simeq \bar{B}^{\mathrm{ch}}_E(\mathcal{F}_X)$$

in the completed chiral bar category.

(iv) A chiral cobar quasi-isomorphism

$$\Theta_{\mathrm{Kos}} : \Omega^{\mathrm{ch}}_E C_X \xrightarrow{\simeq} \mathsf{Den}_{\Delta_S,E}.$$

(v) A primitive Lie comparison

$$\Pi_{\mathrm{prim}} : H^0 \operatorname{Prim}_{\mathrm{prot}}(\mathcal{F}_X) \xrightarrow{\sim} \mathcal{A}_{\mathrm{den}}$$

which preserves the Γ_{ind}-grading, the bracket, the real Serre relations, the imaginary simple-root multiplicities and parities, the orthogonality relations, and the completed Hopf pairing.

Theorem 8.8 *(Microscopic Hall–Drinfeld realization criterion).* If a compact Igusa realization datum exists, then

$$H^0 \operatorname{Prim}_{\mathrm{prot}}(\mathcal{F}_X) \cong \mathcal{A}_{\mathrm{den}} = \mathfrak{g}_{\Delta_5}$$

as charge-graded Lie superalgebras, and

$$\Omega_E^{\mathrm{ch}} C_X \simeq \mathsf{Den}_{\Delta_5,E}.$$

Consequently

$$\operatorname{sdim} H^0 \operatorname{Prim}_{\mathrm{prot}}(\mathcal{F}_X)_{(n,l,m)} = f(nm, l)$$

on Γ_{act}, and the primitive denominator is $64^{-1}\Delta_5(2Z)$.

Conversely, any compact $K3 \times E$ BPS factorization algebra whose primitive Lie algebra is $\mathfrak{g}_{\Delta_5}$ and whose chiral Koszul realization is $\mathsf{Den}_{\Delta_5,E}$ determines the five data of Definition 8.7.

Proof. The primitive comparison is an isomorphism of charge-graded Lie superalgebras, so it identifies the Hall/factorization bracket with the Borcherds–Gritsenko–Nikulin bracket. The protected integration compatibility gives the signed multiplicities $f(nm, l)$. Substitution in the Borcherds denominator identity gives $64^{-1}\Delta_5(2Z)$. The chiral statement is exactly the quasi-isomorphism Θ_{Kos}.

The converse is formal. A compact BPS factorization algebra supplies $\mathcal{F}_X$. Its Hall model, if it realizes the BPS bracket, supplies $\mathcal{H}_X$, protected integration, and the orientation character. A chiral Koszul realization supplies the coalgebra C_X and the cobar quasi-isomorphism. The primitive identification supplies Π_{prim}. □

Proposition 8.9 *(Low-degree bracket information).* The denominator algebra determines the following low-degree data. With

$$\alpha(n, l, m) = n\delta_1 + m\delta_2 + (n + m - l)\delta_3,$$

the active rows through height three are

β	γ	(β, β)	$\operatorname{smult}(\beta)$
δ_i	$(1,1,0), (0,1,1), (0,-1,0)$	2	1
$\delta_i + \delta_j$ $(i < j)$	$(1,2,1), (1,0,0), (0,0,1)$	0	10
$2\delta_i + \delta_j$ $(i \neq j)$	six real rows	2	1
$\delta_1 + \delta_2 + \delta_3$	$(1,1,1)$	-6	-64.

After choosing even Chevalley generators e_i, f_i, h_i for the real simple roots,

$$[h_i, e_j] = (\delta_i, \delta_j)e_j, \qquad [e_i, f_j] = \delta_{ij}h_i, \qquad (\operatorname{ad} e_i)^3 e_j = 0 \quad (i \neq j).$$

For $E_{ij} := [e_i, e_j]$,

$$[f_i, E_{ij}] = 2e_j, \qquad [f_j, E_{ij}] = -2e_i.$$

These are denominator-algebra constants. They are not compact Hall constants.

Proof. The table is obtained by writing $\beta = c_1\delta_1 + c_2\delta_2 + c_3\delta_3$ and solving

$$\gamma(\beta) = (c_1, c_1 + c_2 - c_3, c_2).$$

Then $(\beta, \beta) = -2(4nm - l^2)$, and $\mathrm{smult}(\beta) = f(nm, l)$ follows from Theorem 8.2. The displayed bracket constants are the normalized Chevalley, Jacobi and Serre constants for the symmetric Cartan matrix $A = ((\delta_i, \delta_j))$. □

PROPOSITION 8.10 *(The determinant does not determine Hall constants).* The scalar square, the Borcherds product, and active support determine signed multiplicities. They do not determine the compact Hall structure constants.

Proof. The product determines

$$\mathrm{smult}(\alpha) = \dim \mathfrak{g}_{\alpha,\bar{0}} - \dim \mathfrak{g}_{\alpha,\bar{1}}.$$

A graded super vector space with the same signed dimensions gives the same product. The zero bracket on that graded space gives the same product but not the Borcherds bracket, since $[e_i, e_j] \neq 0$ in $\mathcal{A}_{\mathrm{den}}$ for $i \neq j$. A Hall bracket requires compact extension correspondences, orientation data, protected integration, and a basis of protected primitives. None of these is encoded in the determinant. □

Open Problem 8.11 *(Compact Hall and chiral Koszul realization).* Construct the datum $\mathfrak{K}_X$ of Definition 8.7. Equivalently, construct the compact observable factorization algebra, the oriented HN-completed Hall object, protected integration, the conilpotent chiral bar coalgebra, the chiral cobar quasi-isomorphism to $\mathsf{Den}_{\Delta_s,E}$, and the primitive Lie comparison with $\mathfrak{g}_{\Delta_s}$. The scalar square and the denominator algebra are inputs to this problem. They are not its solution.

Definition 8.12 *(Imaginary simple roots).* For $a \in \Lambda^{2,1}_{II} \cap \mathbb{R}_{>0}\mathcal{P}_{II}$ set

$$\mu_{\bar{0}}(a) = \begin{cases} m(a), & (a,a) < 0 \text{ and } m(a) > 0, \\ \tau(a), & (a,a) = 0 \text{ and } \tau(a) > 0, \\ 0, & \text{otherwise}, \end{cases}$$

and

$$\mu_{\bar{1}}(a) = \begin{cases} -m(a), & (a,a) < 0 \text{ and } m(a) < 0, \\ -\tau(a), & (a,a) = 0 \text{ and } \tau(a) < 0, \\ 0, & \text{otherwise}. \end{cases}$$

This sign split is the Gritsenko–Nikulin BKM parity datum. The product statement alone fixes only a Grothendieck representative for the signed exponent. Define multisets by multiplicity:

$$a \in \Delta^{im}_{\bar{0}} \quad \text{with multiplicity } \mu_{\bar{0}}(a),$$

$$a \in \Delta^{im}_{\bar{1}} \quad \text{with multiplicity } \mu_{\bar{1}}(a).$$

Put

$$\Delta^{im} = \Delta^{im}_{\bar{0}} \cup \Delta^{im}_{\bar{1}}.$$

$$O^{im}_{\mathrm{BPS}} = \bigoplus_{a \in \Delta^{im}} (\mathfrak{g}_{a,\bar{0}} \ominus \mathfrak{g}_{a,\bar{1}}).$$

The real simple roots are

$$\Delta^{re}_{\bar{0}} = \Delta^{re} = \mathcal{P}_{II,prim} = \{\delta_1, \delta_2, \delta_3\}, \qquad \Delta^{re}_{\bar{1}} = \varnothing.$$

Put

$$\Delta_{\bar{p}} = \Delta^{re}_{\bar{p}} \cup \Delta^{im}_{\bar{p}} \qquad (p = 0, 1), \qquad \Delta = \Delta_{\bar{0}} \cup \Delta_{\bar{1}}.$$

The Cartan data satisfy $(\alpha, \alpha) > 0$ for $\alpha \in \Delta^{re}$, $(\alpha, \alpha) \leq 0$ for $\alpha \in \Delta^{im}$, and $(\alpha, \alpha') \leq 0$ for all distinct $\alpha, \alpha' \in \Delta$. If $(\alpha, \alpha) > 0$, then

$$2\frac{(\alpha, \alpha')}{(\alpha, \alpha)} \in \mathbb{Z},$$

since $(\alpha, \alpha) = 2$ and $\Lambda^{2,1}_{II} = \mathbb{Z}\delta_1 + \mathbb{Z}\delta_2 + \mathbb{Z}\delta_3$.

Definition 8.13 *(Generators and relations).* Let I be an index set equipped with a parity map $p : I \to \mathbb{Z}/2\mathbb{Z}$ and a root map $i \mapsto \alpha_i \in \Delta$, with fibres over an imaginary root of cardinality $\mu_{\bar{p}}(a)$ in parity p. If $i \neq j$ and $\alpha_i = \alpha_j$, the two indices are distinct simple roots.

$$\mathrm{ch} : I \to \Lambda^{2,1}_{II}, \qquad i \mapsto \alpha_i.$$

The superalgebra $\mathfrak{g}_{\Delta_S}$ is generated by

$$h_i,\ e_i,\ f_i, \qquad i \in I.$$

The Cartan subalgebra is identified with $\Lambda^{2,1}_{II} \otimes \mathbb{R}$ by $h_i \mapsto \alpha_i$. Relations:

- $[h_i, e_j] = (\alpha_i, \alpha_j)e_j$ and $[h_i, f_j] = -(\alpha_i, \alpha_j)f_j$
- $[e_i, f_j] = \delta_{ij}h_i$
- if $\alpha_i \in \Delta^{re}$ and $i \neq j$, then

$$(\mathrm{ad}\, e_i)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)} e_j = (\mathrm{ad}\, f_i)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)} f_j = 0$$

- if $(\alpha_i, \alpha_j) = 0$, then $[e_i, e_j] = [f_i, f_j] = 0$

The grading is by $\Lambda^{2,1}_{II}$. Put

$$Q_+ = \sum_{\alpha \in \Delta} \mathbb{Z}_{\geq 0}\alpha \subset \Lambda^{2,1}_{II}.$$

The triangular decomposition is

$$\mathfrak{g}_{\Delta_S} = \left(\bigoplus_{\alpha \in Q_+ \setminus \{0\}} \mathfrak{g}_\alpha \right) \oplus (\Lambda^{2,1}_{II} \otimes \mathbb{R}) \oplus \left(\bigoplus_{\alpha \in Q_+ \setminus \{0\}} \mathfrak{g}_{-\alpha} \right).$$

For $i \in I$,

$$\deg e_i = \alpha_i, \qquad \deg f_i = -\alpha_i, \qquad \deg h_i = 0.$$

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}.$$

$$\mathfrak{g}_0 = \Lambda^{2,1}_{II} \otimes \mathbb{R}.$$

The positive and negative roots are

$$\mathcal{R}_+ = \{\alpha \in Q_+ \setminus \{0\} \mid \mathfrak{g}_\alpha \neq 0\}, \qquad \mathcal{R}_- = -\mathcal{R}_+.$$

For every root α, set

$$\mathrm{mult}_{\bar{0}}(\alpha) = \dim \mathfrak{g}_{\alpha,\bar{0}}, \qquad \mathrm{mult}_{\bar{1}}(\alpha) = \dim \mathfrak{g}_{\alpha,\bar{1}}.$$

The signed root superdimension is

$$\mathrm{smult}(\alpha) = \mathrm{mult}_{\bar{0}}(\alpha) - \mathrm{mult}_{\bar{1}}(\alpha).$$

The Weyl–Kac–Borcherds denominator is

$$\begin{aligned}\Phi_{\Delta_5}(z) &= \sum_{w\in W^{(2)}(\Lambda_{II}^{2,1})} \det(w)\left(e^{-2\pi i(w(\rho),z)} - \sum_{a\in\Lambda_{II}^{2,1}\cap\mathbb{R}_{>0}\mathcal{P}_{II}} m(a)e^{-2\pi i(w(\rho+a),z)}\right)\\ &= e^{-2\pi i(\rho,z)} \prod_{\alpha\in\mathcal{R}_+}\left(1-e^{-2\pi i(\alpha,z)}\right)^{\mathrm{smult}(\alpha)}.\end{aligned}$$

$$\Omega(\Lambda_{II}^{2,1}) = \Lambda_{II}^{2,1}\otimes\mathbb{R} + iC(\Lambda_{II}^{2,1})_+.$$

THEOREM 8.14 *(Denominator identity).*

$$\frac{1}{64}\Delta_5(2Z) = \Phi_{\Delta_5}(z).$$

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = \frac{1}{64}\Delta_5(2Z).$$

Proof.

$$\frac{1}{64}\Delta_5(Z) = \sum_{w\in W^{(2)}(\Lambda_{II}^{2,1})} \det(w)\left(e^{-\pi i(w(\rho),z)} - \sum_{a\in\Lambda_{II}^{2,1}\cap\mathbb{R}_{>0}\mathcal{P}_{II}} m(a)e^{-\pi i(w(\rho+a),z)}\right).$$

$$Z\mapsto 2Z: \qquad e^{-\pi i(w(\rho+a),z)} \mapsto e^{-2\pi i(w(\rho+a),z)}.$$

Weyl–Kac–Borcherds:

$$e^{-2\pi i(\rho,z)} \prod_{\alpha\in\mathcal{R}_+}\left(1-e^{-2\pi i(\alpha,z)}\right)^{\mathrm{smult}(\alpha)}.$$

□

PROPOSITION 8.15 *(Gram-index cone and denominator cone).*

$$\alpha(\Gamma_{\mathrm{act}}) = \{\alpha(n,l,m) \mid (n,l,m)\in\Gamma_{\mathrm{eff}},\ f(nm,l)\neq 0\}\subset\Lambda_{II}^{2,1}.$$

$$\alpha(\Gamma_{\mathrm{act}}) \subset \mathcal{R}_+^{\mathrm{vis}} := \{\alpha\in\mathcal{R}_+ \mid \mathrm{smult}(\alpha)\neq 0\}.$$

$$\mathrm{smult}(\alpha(n,l,m)) = \dim\mathfrak{g}_{\alpha(n,l,m),\bar{0}} - \dim\mathfrak{g}_{\alpha(n,l,m),\bar{1}} = f(nm,l) \qquad ((n,l,m)\in\Gamma_{\mathrm{act}}).$$

Proof.

$$\alpha:\Gamma_{\mathrm{ind}}\to\Lambda_{II}^{2,1}, \qquad (n,l,m)\mapsto 2nf_2 - lf_3 + 2mf_{-2}.$$

$$q^n r^l s^m = \exp(-\pi i(\alpha(n,l,m),z))$$

$$Z\mapsto 2Z.$$

$$\frac{1}{64}\Delta_5(2Z) = e^{-2\pi i(\rho,z)} \prod_{\alpha\in\mathcal{R}_+} \left(1 - e^{-2\pi i(\alpha,z)}\right)^{\mathrm{smult}(\alpha)}.$$

$$\frac{1}{64}\mathcal{D}_X(2Z) = e^{-2\pi i(\rho,z)} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} \left(1 - e^{-2\pi i(\alpha(n,l,m),z)}\right)^{f(nm,l)}.$$

The triples in $\Gamma_{\mathrm{eff}} \setminus \Gamma_{\mathrm{act}}$ have exponent zero and do not contribute visible product factors. The denominator product alone does not prove absence of root spaces with zero signed superdimension.

$$\mathcal{D}_X = \Delta_5 \quad \Longrightarrow \quad \mathrm{smult}(\alpha(n,l,m)) = f(nm,l) \quad ((n,l,m) \in \Gamma_{\mathrm{act}}).$$

□

Proposition 8.16 *(What the denominator determines).* The identity

$$\frac{1}{64}\Delta_5(2Z) = e^{-2\pi i(\rho,z)} \prod_{\alpha\in\mathcal{R}_+} \left(1 - e^{-2\pi i(\alpha,z)}\right)^{\mathrm{smult}(\alpha)}$$

determines the signed integers $\mathrm{smult}(\alpha)$. It does not determine the two parity dimensions

$$\mathrm{mult}_{\bar{0}}(\alpha), \qquad \mathrm{mult}_{\bar{1}}(\alpha),$$

and it does not determine the structure constants

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \to \mathfrak{g}_{\alpha+\beta}.$$

Proof. Taking the logarithm of the product reads off only the exponent attached to each monomial. That exponent is

$$\mathrm{smult}(\alpha) = \mathrm{mult}_{\bar{0}}(\alpha) - \mathrm{mult}_{\bar{1}}(\alpha).$$

The difference $a - b$ does not determine the pair (a, b), and no bilinear bracket appears in the product. □

Proposition 8.17 *(Index equality is not a state-space identification).* The equality

$$\mathrm{smult}(\alpha(n,l,m)) = f(nm,l)$$

identifies the root-space Euler characteristic of $\mathcal{A}_{\mathrm{den}}$ with the virtual K3 half-index coefficient. It does not prove an isomorphism

$$\mathfrak{g}_{\alpha(n,l,m)} \cong \mathcal{H}^{\mathrm{BPS}}_{(n,l,m)}$$

and it does not prove that the microscopic operator product on BPS states is the Borcherds bracket.

Theorem 8.18 *(What the determinant forgets).* Let

$$\mathcal{B} = \bigoplus_{\gamma\in\Gamma_{\mathrm{eff}}} \mathcal{B}_\gamma$$

be any charge-graded super vector space with finite-dimensional homogeneous pieces and

$$\mathrm{sdim}\, \mathcal{B}_{(n,l,m)} = f(nm,l).$$

Then its Gram-graded Euler/Fock determinant is

$$64q^{1/2}r^{1/2}s^{1/2}\prod_{\gamma\in\Gamma_{\mathrm{eff}}}(1-x^{\gamma})^{\operatorname{sdim}\mathcal{B}_{\gamma}}=\Delta_5(Z).$$

If $\mathcal{B}$ carries a bracket compatible with charge addition, the determinant forgets that bracket and remembers exactly the class of $\mathcal{B}$ in

$$K_0(\Gamma_{\mathrm{eff}}\text{-graded finite super vector spaces}).$$

Thus the determinant of $\mathcal{A}_{\mathrm{den}}=\mathfrak{g}_{\Delta_5}$ is the Grothendieck shadow of its denominator bracket. The bracket exists, but the determinant sees only its charge-graded Grothendieck class.

Proof. The Fock determinant lemma gives

$$\operatorname{sdet}_{\mathcal{B}}(1-x)=\prod_{\gamma\in\Gamma_{\mathrm{eff}}}(1-x^{\gamma})^{\operatorname{sdim}\mathcal{B}_{\gamma}}.$$

The hypothesis $\operatorname{sdim}\mathcal{B}_{(n,l,m)}=f(nm,l)$ identifies this product with the Borcherds product for $64^{-1}\Delta_5$. Multiplying by the vacuum factor gives Δ_5. A bracket

$$[\mathcal{B}_{\gamma},\mathcal{B}_{\gamma'}]\to\mathcal{B}_{\gamma+\gamma'}$$

does not occur in the determinant formula. The determinant is therefore invariant under replacing $\mathcal{B}$ by any other graded super vector space with the same Grothendieck class. □

9 The Jacobi coefficients as superdimensions

$$\operatorname{smult}(\alpha(n,l,m))=f(nm,l).$$

Eichler–Zagier normalization [12]:

$$Z_{\mathrm{K3}}=2\phi_{0,1}.$$

Let $\phi_{12,1}$ be the first Fourier–Jacobi coefficient of Δ_{12}. Its integral coprime expansion begins

$$\phi_{12,1}(z_1,z_2)=(r^{-1}+10+r)q+(10r^{-2}-88r^{-1}-132-88r+10r^2)q^2+\dots$$

Let

$$\delta_{12}(z_1)=q\prod_{n\geq 1}(1-q^n)^{24}.$$

$$\phi_{0,1}(z_1,z_2)=\frac{\phi_{12,1}}{\delta_{12}}=\sum_{\substack{n\geq 0\\ l\in\mathbb{Z}}}f(n,l)\exp(2\pi i(nz_1+lz_2)).$$

$$\phi_{0,1}\in J_{0,1}^{\mathrm{weak}},$$

$$f(n,l)=0\quad\text{unless}\quad n\geq 0,\quad 4n-l^2\geq -1,$$

$$f(n,l)=f(n,-l).$$

$$\phi_{0,1}(z_1,z_2)=(r^{-1}+10+r)+q(10r^{-2}-64r^{-1}+108-64r+10r^2)+\dots$$

Theorem 9.1 *(Borcherds product).*

$$\frac{1}{64}\Delta_5(Z) = \exp(\pi i(z_1+z_2+z_3)) \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1-\exp(2\pi i(nz_1+lz_2+mz_3)))^{f(nm,l)}.$$

Remark 9.2.

$$\begin{aligned}\Gamma_{\mathrm{eff}} = \{(n,l,m) \mid m>0,\ n\geq 0,\ l\in\mathbb{Z}\}\\ \cup\{(n,l,0) \mid n>0,\ l\in\mathbb{Z}\}\\ \cup\{(0,l,0) \mid l<0\}.\end{aligned}$$

Proof.

$$f(n,l) = O(\exp(\sqrt{4n-l^2})).$$

The Borcherds product theorem, in the Gritsenko–Nikulin normalization, applies to the input $\phi_{0,1}$ and gives a meromorphic Siegel modular form convergent in the zero-dimensional cusp chamber [5, Theorem 13.3], [19, Section 2].

$$P(Z) := \exp(\pi i(z_1+z_2+z_3)) \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1-e^{2\pi i(nz_1+lz_2+mz_3)})^{f(nm,l)}.$$

$$\exp(\pi i(z_1+z_2+z_3)) \prod_{\substack{n>0,\ l\in\mathbb{Z}\\ \text{or } n=0,\ l<0}} \left(1-e^{2\pi i(nz_1+lz_2)}\right)^{f(0,l)}$$

$$\times \prod_{\substack{n\geq 0,\ m>0\\ l\in\mathbb{Z}}} \left(1-e^{2\pi i(nz_1+lz_2+mz_3)}\right)^{f(nm,l)},$$

with

$$f(0,0)=10, \qquad f(0,-1)=1, \qquad f(0,l)=0 \quad (l<-1).$$

$$\log(1-X) = -\sum_{k\geq 1} X^k/k,$$

$$\log\left(\prod_{\substack{n\geq 0,\ m>0\\ l\in\mathbb{Z}}} (1-q^n r^l s^m)^{f(nm,l)}\right) = -\sum_{k\geq 1}\frac{1}{k}\sum_{\substack{n\geq 0,\ m>0\\ l\in\mathbb{Z}}} f(nm,l) q^{kn} r^{kl} s^{km}.$$

$$[s^d]\log P = -\sum_{k\mid d}\frac{1}{k}\sum_{\substack{n\geq 0\\ l\in\mathbb{Z}}} f\left(n\frac{d}{k},l\right) q^{kn} r^{kl}.$$

$$P^{-1} = q^{-1/2} r^{-1/2} s^{-1/2} \operatorname{sch}\mathcal{F}(\mathcal{V}).$$

$$P|_5\Gamma_\infty = \nu_\infty P.$$

$$\psi_{5,\frac{1}{2}}(z_1,z_2)e^{\pi i z_3}.$$

For $\gamma\in\Gamma_\infty/\{\pm\mathbf{I}_4\}$,

$$P|_5\gamma = \nu_\infty(\gamma)P, \qquad \nu_\infty : \Gamma_\infty/\pm\mathbf{I}_4 \to \{\pm 1\}.$$

For

$$S = \begin{pmatrix} 0 & 1 & & 0 \\ 1 & 0 & & \\ & & 0 & 1 \\ 0 & & 1 & 0 \end{pmatrix},$$

$$P|_5 S = \nu_{\Delta_5}(S) P.$$

$$\langle \Gamma_\infty, S \rangle = \mathbf{PSp}_4(\mathbb{Z}), \qquad P \in M_5(\mathbf{Sp}_4(\mathbb{Z}), \nu_{\Delta_5}).$$

$$[q^{1/2} r^{1/2} s^{1/2}]\, 64P = 64.$$

□

THEOREM 9.3 *($N = 1$ determinant–denominator square root).* Set

$$q = \exp(2\pi i z_1), \quad r = \exp(2\pi i z_2), \quad s = \exp(2\pi i z_3).$$

$$\phi_{0,1}(\tau, z) = 4\left(\frac{\vartheta_2(\tau,z)^2}{\vartheta_2(\tau,0)^2} + \frac{\vartheta_3(\tau,z)^2}{\vartheta_3(\tau,0)^2} + \frac{\vartheta_4(\tau,z)^2}{\vartheta_4(\tau,0)^2}\right).$$

$$\phi_{0,1} = (r^{-1} + 10 + r) + q(10r^{-2} - 64r^{-1} + 108 - 64r + 10r^2)$$
$$+q^2(r^{-3} + 108r^{-2} - 513r^{-1} + 808 - 513r + 108r^2 + r^3) + O(q^3).$$

$$f(0,0) = 10, \qquad \mathrm{wt}\, \Delta_5 = f(0,0)/2 = 5.$$

$$D(Z) := \frac{1}{64}\Delta_5(Z) = q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

$$\Delta_{10}(Z) = \Delta_5(Z)^2 = 4096\, qrs \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{2f(nm,l)}.$$

$$Z^X_\square = C_\square / \Delta_5^2, \qquad C_\square \in \mathbb{C}^\times.$$

$$D_X(Z) := \Delta_5(Z) = 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

$$D_X(Z)^2 = \frac{C_\square}{Z^X_\square}.$$

$$\frac{1}{64}\Delta_5(2Z) = \Phi_{\Delta_5}(z).$$

$$\mathcal{Z}^{\mathrm{vir}}_{\mathrm{BPS}} = D_X^{-1}$$

$$\left(\mathcal{Z}^{\mathrm{vir}}_{\mathrm{BPS}}\right)^2 = Z^X_\square / C_\square.$$

$$\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2},$$

$$f(nm,l) = \dim \mathfrak{g}_{\alpha(n,l,m),\bar{0}} - \dim \mathfrak{g}_{\alpha(n,l,m),\bar{1}}, \qquad q^n r^l s^m = \exp(-\pi i(\alpha(n,l,m), z))$$

for $(n,l,m) \in \Gamma_{\mathrm{act}}$.

Proof.

$$\phi_{0,1} \in J_{0,1}^{\mathrm{weak}}, \qquad Z_{\mathrm{K3}} = 2\phi_{0,1}.$$

$$\mathrm{Borch}(\phi_{0,1}) = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

$$\mathrm{Borch}(\phi_{0,1}) = \Delta_5 \qquad \text{(Borcherds–Gritsenko–Nikulin).}$$

Borcherds–Gritsenko–Nikulin: [5], [19].

$$\Delta_{10} = \Delta_5^2.$$

$$Z_{\square}^X = C_{\square}/\Delta_5^2,$$

$$\frac{C_{\square}}{Z_{\square}^X} = \mathrm{Borch}(\phi_{0,1})^2.$$

$$\mathrm{sdim}\, \mathcal{V}_{n,l,m} = f(nm, l)$$

$$\mathrm{sch}\, \mathcal{F}(\mathcal{V}) = \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{-f(nm,l)}.$$

$$\mathcal{Z}_{\mathrm{BPS}}^{\mathrm{vir}} = \frac{1}{64} q^{-1/2} r^{-1/2} s^{-1/2}\, \mathrm{sch}\, \mathcal{F}(\mathcal{V}).$$

$$\left(\mathcal{Z}_{\mathrm{BPS}}^{\mathrm{vir}}\right)^{-1} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

□

10 Eight diagonal-divisor factorization rows

$$g = 2.$$

$$t, N \in \mathbb{Z}_+ : \quad \Gamma_t(N) = \left\{ \begin{pmatrix} * & *t & * & * \\ & * & * & *t^{-1} \\ N & *Nt & * & * \\ Nt & *Nt & *t & * \end{pmatrix} \in \mathbf{Sp}_4(\mathbb{Q}) \mid * \in \mathbb{Z} \right\}.$$

$$\mathbb{H}_2/\Gamma_t \longleftrightarrow \{(1,t)\text{-polarized abelian surfaces}\}.$$

$$\mathcal{H}_{\mathrm{diag}} = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mid a, b \in \mathbb{H}_1 \right\} \subset \mathbb{H}_2.$$

Theorem 10.1 *(Clery–Gritsenko).* In the class of Siegel modular forms with character or multiplier system for the Hecke-type congruence subgroups $\Gamma_t(N)$ whose divisor is the union of the $\Gamma_t(N)$-translates of $\mathcal{H}_{\mathrm{diag}}$ with multiplicity one, there are exactly eight rows [18, Theorem 1.4]. The nine-candidate list is [18, Proposition 1.2]; the proof of Theorem 1.4 excludes only the ninth candidate $(t, N; k) = (2, 4; \frac{1}{2})$. The congruence Borcherds product with all cusp coefficients is [18, Theorem 3.1], with the final Section 3 product displays for ∇_3, ∇_2, $\nabla_{3/2}$, and Q_1.

j	F_j	Γ	$\mathrm{wt}(F_j)$	character
1	Δ_5	Γ_1	5	$v_\eta^{12} \times v_H$
2	Δ_2	Γ_2^+	2	$v_\eta^{6} \times v_H$
3	Δ_1	Γ_3^+	1	$v_\eta^{4} \times v_H$
4	$\Delta_{1/2}$	Γ_4^+	$\frac{1}{2}$	$v_\eta^{3} \times v_H$ (degree 8)
5	∇_3	$\Gamma_0^{(2)}(2)$	3	$\chi_2^{(2)} \times v_H$
6	∇_2	$\Gamma_0^{(2)}(3)$	2	$\chi_2^{(3)} \times v_H$
7	$\nabla_{3/2}$	$\Gamma_0^{(2)}(4)$	$\frac{3}{2}$	$\chi_4^{(4)} \times v_H$
8	Q_1	$\Gamma_2(2)$	1	$\chi_4^{(2)} \times v_H$

$$\operatorname{supp} \operatorname{div}_{\mathbb{H}_2}(F_j) = \bigcup_{\gamma \in \Gamma_j} \gamma \langle \mathcal{H}_{\mathrm{diag}} \rangle, \qquad \mathrm{mult}_{\mathcal{H}_{\mathrm{diag}}}(F_j) = 1.$$

$$F_1 = \Delta_5, \qquad \frac{1}{64} F_1(2Z) = \mathrm{den}(\mathfrak{g}_{\Delta_5}).$$

By Theorem B.7 the full-paramodular rows F_2, F_3, F_4 also carry Gritsenko–Nikulin denominator identities. The congruence rows F_5, F_6, F_7, F_8 have product-degree lattices in Proposition B.9. A nonabelian Weyl–Borcherds promotion is additional structure: it is the CHL denominator theorem of Govindarajan for F_5, F_6, F_7, F_8. For F_7 the theorem lives on the μ_4-multiplier cover determined by the Clery–Gritsenko order-four multiplier, with no Vol. III CY/DT/chiral boundary host.

PROPOSITION 10.2 *(Extension separation).* No statement in this section is used in the $N = 1$ theorem for Δ_5. For every row the product formula itself gives a charged primitive product. It gives a Gram-graded Euler/Fock determinant only after a one-particle index and a charge-invariant interpretation have been supplied. A nonabelian denominator theorem is stronger: it additionally requires a Weyl chamber, real simple roots, a reflection character, and a Weyl alternating sum.

Conjecture 10.3 *(Denominator extension beyond known rows).* For every row not already covered by the Gritsenko–Nikulin denominator theorems or by the Govindarajan CHL/WKB denominator theorem, whose datum

$$\mathscr{P}_j = (\phi_j, \Gamma^+_{j,\mathrm{prod}}, \Gamma_{j,\mathrm{mod}}, \rho_j, \nu_j, L_j, \alpha_j)$$

has been constructed, with

$$\phi^{(j)}(\tau, z) = \sum c_j(n, l) q^n r^l,$$

there is a positive product cone $\Gamma^+_{j,\mathrm{prod}}$, a Weyl vector ρ_j, and a generalized Borcherds–Kac–Moody superalgebra $\mathfrak{g}_j$ such that

$$F_j(Z) = C_j e^{2\pi i(\rho_j, Z)} \prod_{\gamma \in \Gamma^+_{j,\mathrm{prod}}} (1 - e^{2\pi i(\gamma, Z)})^{c_j(\gamma)},$$

$$\mathrm{den}(\mathfrak{g}_j) = C'_j F_j(a_j Z), \qquad \mathrm{sdim}(\mathfrak{g}_{j,\gamma}) = c_j(\gamma).$$

If a CHL/K3 sector supplies the row with

$$\phi^{(j)} = \tfrac{1}{2} Z^{(j)}_{\mathrm{K3}},$$

then the scalar-square branch is

$$Z_j = C_j / F_j^2.$$

Proposition 10.4 *(CHL squares and denominator signs).* The identities

$$\nabla_3^2 = \Phi_6, \qquad \nabla_2^2 = \Phi_4$$

which occur in the CHL dyon literature do not by themselves give denominator identities for the square-root rows $F_5 = \nabla_3$ and $F_6 = \nabla_2$. They give scalar Siegel modular forms on the CHL side. To descend from a scalar square to a denominator theorem for a root form, one must also construct the reflection character, the parity, and the Weyl anti-invariant additive side. Govindarajan's twisted CHL theorem supplies exactly this missing structure for

$$F_5 = \nabla_3, \qquad F_6 = \nabla_2, \qquad F_7 = \nabla_{3/2}, \qquad F_8 = Q_1,$$

as recorded in Theorem B.13.

Proof. Clery–Gritsenko record the dd-forms ∇_3 and ∇_2 and note that their squares were constructed in the CHL papers as Φ_6 and Φ_4. The square forgets the sign character which a denominator identity needs on simple real reflections. Thus the scalar CHL product is evidence for the automorphic product, not by itself a proof of the BKM datum $(P_j, \mathcal{M}_j, W_j, \epsilon_j, \rho_j, N_j(a))$ for the root form. The additional datum is supplied by [14], building on [15] and [8]. □

The Vol. III square-root comparison tuple has weights

$$(5, 2, 1, 1, \tfrac{1}{2}, 1, \tfrac{1}{4}, 0).$$

It is not the Clery–Gritsenko table. A comparison with the Clery–Gritsenko forms requires a row map. Thus

$$\text{Clery–Gritsenko classification} \neq \text{CHL realisation theorem.}$$

$$F_1 : \text{ theorem}, \qquad F_2, F_3, F_4 : \text{ GN denominator theorem},$$

$$F_5, F_6, F_7, F_8 : \text{ Govindarajan CHL/WKB denominator theorem},$$

with F_7 understood on the μ_4-multiplier cover determined by its order-four multiplier and not as a Vol. III physical boundary theorem. Dijkgraaf–Verlinde–Verlinde: [11]. Borcherds: [5].

A Spin L-factor normalisation for Δ_5^2

Set the theta-constant normalisation

$$\Delta_{10}^{\theta} := \Delta_5^2.$$

The Oberdieck–Pixton monic normalisation is

$$\chi_{10}^{\mathrm{OP}} := 2^{-12}\Delta_{10}^{\theta} = 4096^{-1}\Delta_5^2.$$

The spinor L-function is attached to the Hecke eigenline

$$\mathbb{C}\Delta_{10}^{\theta} = \mathbb{C}\chi_{10}^{\mathrm{OP}};$$

it is independent of the non-zero scalar used to name its generator. Put

$$\Delta(q) = q\prod_{n\geq 1}(1-q^n)^{24}, \qquad E_6(q) = 1 - 504\sum_{n\geq 1}\sigma_5(n)q^n,$$

$$g := \Delta E_6 = \sum_{n\geq 1} a_n(g)q^n \in S_{18}(\mathbf{SL}_2(\mathbb{Z})).$$

THEOREM A.1 *(Andrianov factorisation).* For the Saito–Kurokawa representation generated by Δ_{10}^{θ},

$$L_{\mathrm{spin}}(s, \Delta_{10}^{\theta}) = \zeta(s-8)\zeta(s-9)L(s, g).$$

Here L_{spin} is Andrianov's uncompleted classical spinor L-function. For a genus-two form of weight 10, the completed function is

$$L_{\mathrm{spin}}^{*}(s, F) = (2\pi)^{-2s}\Gamma(s)\Gamma(s-8)L_{\mathrm{spin}}(s, F).$$

Proof. The Igusa cusp form Δ_{10}^{θ} spans the weight-10 Maass Spezialschar line attached by the Saito–Kurokawa correspondence to the unique normalized form $g = \Delta E_6 \in S_{18}(\mathbf{SL}_2(\mathbb{Z}))$ [36]. This is Andrianov's spinor L-factor formula for the degree-2 Saito–Kurokawa lift attached to the weight-18 elliptic cusp form g [1]. Since $\dim S_{18}(\mathbf{SL}_2(\mathbb{Z})) = 1$, the source form is ΔE_6. □

PROPOSITION A.2 *(Central-zero cancellation).* With

$$\Lambda(g, s) := (2\pi)^{-s}\Gamma(s)L(s, g),$$

one has

$$\Lambda(g, s) = -\Lambda(g, 18-s), \qquad L(9, g) = 0.$$

Hence the possible pole of $\zeta(s-8)L(s, g)$ at $s = 9$ is cancelled. At $s = 10$ the residue is

$$\operatorname*{Res}_{s=10} L_{\mathrm{spin}}(s, \Delta_{10}^{\theta}) = \zeta(2)L(10, g) = \frac{\pi^2}{6}L(10, g).$$

Moreover $L(10, g) \neq 0$. Thus $L_{\mathrm{spin}}(s, \Delta_{10}^{\theta})$ has a simple pole at $s = 10$ with non-zero residue.

Proof. For a level-one cusp form of weight 18, the root number is $(-1)^{18/2} = -1$. Thus the completed L-function is odd about $s = 9$, and the central value vanishes. The factor $\zeta(s-8)$ has its only possible pole at $s = 9$, so the product $\zeta(s-8)L(s, g)$ is regular there. At $s = 10$ the factor $\zeta(s-9)$ has residue 1, while $\zeta(s-8) = \zeta(2)$. Write

$$L_p(s, g) = (1 - \alpha_p p^{-s})^{-1}(1 - \beta_p p^{-s})^{-1}.$$

Deligne's Ramanujan–Petersson theorem gives

$$|\alpha_p| = |\beta_p| = p^{17/2}.$$

At $s = 10$,

$$|\alpha_p p^{-10}| = |\beta_p p^{-10}| = p^{-3/2},$$

so the Euler product converges absolutely and has no zero. □

PROPOSITION A.3 *(Manin–Deligne period).* There are Manin periods $\Omega^{\pm}(g)$ such that

$$\varepsilon(m) = \begin{cases} +, & m \equiv 0 \pmod 2, \\ -, & m \equiv 1 \pmod 2, \end{cases} \qquad \frac{L(m, g)}{(2\pi i)^m \Omega^{\varepsilon(m)}(g)} \in \mathbb{Q}, \qquad 1 \leq m \leq 17.$$

In particular

$$L(10, g) \in \mathbb{Q}^{\times}(2\pi i)^{10}\Omega^{+}(g).$$

Proof. This is the Eichler–Shimura–Manin period theorem for the rational newform g [26], [24], in Deligne's period normalisation [9]. The nonvanishing at $m = 10$ is supplied by the Euler product in Proposition A.2. No identity of the form

$$L(10, g) = q\, a_{10}(g)\Omega^-(g), \qquad q \in \mathbb{Q}^\times,$$

is used. □

Proposition A.4 *(Coefficient check).*

$$a_{10}(g) = 541648800 = 2^5 \cdot 3^2 \cdot 5^2 \cdot 7 \cdot 11 \cdot 977.$$

Proof.

$$a_{10}(g) = \tau(10) - 504 \sum_{k=1}^{9} \tau(k)\sigma_5(10-k).$$

$$(\tau(1), \ldots, \tau(10)) = (1, -24, 252, -1472, 4830, -6048, -16744, 84480, -113643, -115920),$$

$$(\sigma_5(1), \ldots, \sigma_5(9)) = (1, 33, 244, 1057, 3126, 8052, 16808, 33825, 59293).$$

Substitution gives $a_{10}(g) = 541648800$. Equivalently, since g is a Hecke eigenform,

$$a_{10}(g) = a_2(g)a_5(g) = (-528)(-1025850) = 541648800.$$

□

Remark A.5 *(Scalar-square separations).*

claim	failure	correct statement
$\chi_{10}^{\mathrm{OP}} = \Delta_5^2$	normalisation conflict	$\chi_{10}^{\mathrm{OP}} = 2^{-12}\Delta_5^2$
$L_{\mathrm{spin}} = \zeta(s-8)\zeta(s-9)L(s,g)$	scalar normalisation	Hecke eigenline
unique pole from $\zeta(s-9)$	$\zeta(s-8)$ also has $s = 9$	$L(9, g) = 0$
$L(10, g) \sim a_{10}(g)\Omega^-(g)$	wrong period parity	$L(10, g) \sim \Omega^+(g)$
arithmetic shadow $\Rightarrow$ BKM proof	wrong implication	appendix only

B Eight diagonal-divisor boundary rows

Let F_j denote the diagonal-divisor forms in the present F-order. The Clery–Gritsenko source order is

$$\Delta_5, \Delta_2, \nabla_3, \Delta_1, \nabla_2, \Delta_{1/2}, \nabla_{3/2}, Q_1.$$

The present order is [18]

j	F_j	(t, N)	Γ_j	wt
1	Δ_5	$(1, 1)$	Γ_1	5
2	Δ_2	$(2, 1)$	Γ_2^+	2
3	Δ_1	$(3, 1)$	Γ_3^+	1
4	$\Delta_{1/2}$	$(4, 1)$	Γ_4^+	$\frac{1}{2}$
5	∇_3	$(1, 2)$	$\Gamma_0^{(2)}(2)$	3
6	∇_2	$(1, 3)$	$\Gamma_0^{(2)}(3)$	2
7	$\nabla_{3/2}$	$(1, 4)$	$\Gamma_0^{(2)}(4)$	$\frac{3}{2}$
8	Q_1	$(2, 2)$	$\Gamma_2(2)$	1

and

$$\operatorname{supp}\operatorname{div}_{\mathbb{H}_2}(F_j) = \bigcup_{\gamma\in\Gamma_j} \gamma\langle \mathcal{H}_{\mathrm{diag}}\rangle, \qquad \operatorname{mult}_{\mathcal{H}_{\mathrm{diag}}}(F_j) = 1.$$

Here $F_7 = \nabla_{3/2}$ is the Clery–Gritsenko row at $(t, N) = (1, 4)$. It is not the Vol. III physical $\Phi_7^{\mathrm{V}_3}$ slot until a separate row map has been constructed.

The characters and multiplier systems are

j	ν_j
1	$v_\eta^{12} \times v_H$
2	$v_\eta^{6} \times v_H$
3	$v_\eta^{4} \times v_H$
4	$v_\eta^{3} \times v_H$ (degree 8)
5	$\chi_2^{(2)} \times v_H$
6	$\chi_2^{(3)} \times v_H$
7	$\chi_4^{(4)} \times v_H$ (order 4)
8	$\chi_4^{(2)} \times v_H$

Definition B.1 *(Row proof datum).* For a row j, a denominator proof consists of

$$\mathscr{P}_j = (\phi_j, \Gamma^+_{j,\mathrm{prod}}, \Gamma_{j,\mathrm{mod}}, \rho_j, \nu_j, L_j, \alpha_j),$$

where

$$\phi_j(\tau, z) = \sum c_j(n, l) q^n r^l,$$
$$F_j(Z) = C_j e^{2\pi i(\rho_j, Z)} \prod_{\gamma\in\Gamma^+_{j,\mathrm{prod}}} (1 - e^{2\pi i(\gamma, Z)})^{c_j(\gamma)},$$
$$\nu_j : \Gamma_{j,\mathrm{mod}} \to \mathbb{C}^\times, \qquad \operatorname{supp}\operatorname{div}(F_j) = \Gamma_{j,\mathrm{mod}} \cdot \mathcal{H}_{\mathrm{diag}}, \qquad \operatorname{mult}_{\mathcal{H}_{\mathrm{diag}}}(F_j) = 1,$$

and

$$\alpha_j : \Gamma^+_{j,\mathrm{prod}} \to L_j, \qquad \operatorname{sdim}(\mathfrak{g}_{j,\alpha_j(\gamma)}) = c_j(\gamma).$$

For a product with several cusps this single-source notation is only a template. The congruence-row product datum is the cusp-labelled datum of Proposition B.9.

Definition B.2 *(Gritsenko–Nikulin degree lattice).* For $t \in \mathbb{Z}_{>0}$ set

$$M_t := U(4t) \oplus \langle 2\rangle = \mathbb{Z}f_2 \oplus \mathbb{Z}\widehat{f_3} \oplus \mathbb{Z}f_{-2},$$

with Gram matrix

$$\begin{pmatrix} 0 & 0 & -4t \\ 0 & 2 & 0 \\ -4t & 0 & 0 \end{pmatrix}.$$

For $(n, l, m) \in \mathbb{Z}^3$ put

$$\alpha_t(n, l, m) := nf_2 + l\widehat{f_3} + mf_{-2} \in M_t, \qquad e_t^{(n,l,m)} := q^n r^l s^{tm}.$$

Then

$$(\alpha_t(n, l, m), \alpha_t(n, l, m)) = -2(4tnm - l^2).$$

Let Γ_t^{prod} be the Gritsenko–Nikulin product cone: $(n, l, m) > 0$ if $m > 0$, or if $m = 0, n > 0$, or if $m = n = 0, l < 0$.

Lemma B.3 *(Two exponential normalizations).* Let

$$D_5(Z) := 64^{-1}\Delta_5(Z).$$

Gritsenko–Nikulin's Siegel normalization gives

$$D_5(Z) = q^{1/2}r^{1/2}s^{1/2}\prod_{(n,l,m)>0}(1-q^n r^l s^m)^{f(nm,l)}$$

with leading monomial $q^{1/2}r^{1/2}s^{1/2}$. The denominator normalization used for the main BKM algebra is

$$q^n r^l s^m = \exp(-\pi i(\alpha(n,l,m),z)).$$

Thus replacing Z by $2Z$ converts the product character to the usual $\exp(-2\pi i(\alpha,z))$ denominator character, and

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 64^{-1}\Delta_5(2Z).$$

For rows F_2, F_3, F_4, Appendix B uses the Gritsenko–Nikulin exponential

$$e_t^{(n,l,m)} = q^n r^l s^{tm}$$

directly. Hence

$$C'_j = 1, \qquad a_j = 1 \qquad (j = 2,3,4)$$

is a statement in the e_t normalization. It is not the doubled $N = 1$ normalization of the main Δ_5 denominator.

Proof. The first product is Gritsenko–Nikulin [19, Theorem 4.1]. Their Proposition 3.1 identifies $D_5(2Z)$ with the denominator function of the corrected Lorentzian BKM superalgebra. The rows F_2, F_3, F_4 are the separate $e_t^{(n,l,m)} = q^n r^l s^{tm}$ denominator identities of Gritsenko–Nikulin [20, Theorem 2.1, (2.3), (2.11), Lemma 2.5, (2.18), (2.19), Theorem 2.6, (2.20), (2.21), Section 5.1.1]. □

Proposition B.4 *(Full-paramodular rows).* Let

$$\phi_{0,2} = \sum f_2(n,l)q^n r^l, \qquad \phi_{0,3} = \sum f_3(n,l)q^n r^l, \qquad \phi_{0,4} = \sum f_4(n,l)q^n r^l,$$

where [20]

$$\phi_{0,2} = \frac{1}{2}\eta^{-4}\sum_{m,n\in\mathbb{Z}}(3m-n)\left(\frac{-4}{m}\right)\left(\frac{12}{n}\right)q^{(3m^2+n^2)/24}r^{(m+n)/2},$$

$$\phi_{0,3} = \left(\frac{\vartheta(\tau,2z)}{\vartheta(\tau,z)}\right)^2, \qquad \phi_{0,4} = \frac{\vartheta(\tau,3z)}{\vartheta(\tau,z)}.$$

Then the rows F_2, F_3, F_4 have the following product data:

j	F_j	ϕ_j	L_j	ρ_j
2	Δ_2	$\phi_{0,2}$	M_2	$(\frac{1}{4},\frac{1}{2},\frac{1}{4})$
3	Δ_1	$\phi_{0,3}$	M_3	$(\frac{1}{6},\frac{1}{2},\frac{1}{6})$
4	$\Delta_{1/2}$	$\phi_{0,4}$	M_4	$(\frac{1}{8},\frac{1}{2},\frac{1}{8})$

and

$$\alpha_j^{\mathrm{GN}}(n,l,m) := \alpha_t(n,l,m) \quad \text{for } (j,t) = (2,2),(3,3),(4,4).$$

The product identities are

$$\Delta_2(Z) = q^{1/4} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_2^{\mathrm{prod}}} (1 - q^n r^l s^{2m})^{f_2(nm,l)},$$

$$\Delta_1(Z) = q^{1/6} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_3^{\mathrm{prod}}} (1 - q^n r^l s^{3m})^{f_3(nm,l)},$$

$$\Delta_{1/2}(Z) = q^{1/8} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_4^{\mathrm{prod}}} (1 - q^n r^l s^{4m})^{f_4(nm,l)}.$$

The characters are respectively induced by $v_\eta^6 \times v_H$, $v_\eta^4 \times v_H$, and $v_\eta^3 \times v_H$. Equivalently, on the extended paramodular groups,

$$\nu_2 = \chi_4 : \Gamma_2^+ \to \mu_4, \qquad \nu_3 = \chi_6 : \Gamma_3^+ \to \mu_6, \qquad \nu_4 = v_8 : \Gamma_4^+ \to \mu_8.$$

Thus for $(j,t) = (2,2), (3,3), (4,4)$,

$$\Gamma_{j,\mathrm{prod}}^+ = \Gamma_t^{\mathrm{prod}}, \qquad \Gamma_{j,\mathrm{mod}} = \Gamma_t^+.$$

Proof. The product formulae are Gritsenko–Nikulin [20], Theorem 2.1 and formulae (2.21), (2.20), (2.11). Their Section 5 puts all formulae of Lie type on the hyperbolic lattice M_t and writes the exponent as $e^{(n,l,m)} = q^n r^l s^{tm}$. The Weyl vector is

$$\rho_{t,II,\bar{0}} = \left(\frac{1}{2t}, \frac{1}{2}, \frac{1}{2t}\right),$$

which gives the three displayed prefactors for $t = 2, 3, 4$. □

PROPOSITION B.5 *(Gritsenko–Nikulin chambers for F_2, F_3, F_4).* For $t = 2, 3, 4$ the Weyl group is

$$W_{t,II,\bar{0}} = W^{(2)}(M_t),$$

and all simple real roots in this chamber are even. The Weyl vector is

$$\rho_t = \left(\frac{1}{2t}, \frac{1}{2}, \frac{1}{2t}\right).$$

The simple real roots are

$$P_2 = \{(0,-1,0),\ (1,1,0),\ (1,3,1),\ (0,1,1)\},$$

$$P_3 = \{(0,-1,0),\ (1,1,0),\ (2,5,1),\ (2,7,2),\ (1,5,2),\ (0,1,1)\},$$

and

$$P_4 = \{\delta \in M_4 \mid (\delta,\delta) = 2,\ (\delta,\rho_4) = -1\}.$$

Thus the automorphic denominator chamber for F_j is the chamber in $M_t \otimes \mathbb{R}$ cut out by $(x,\delta) \le 0$ for $\delta \in P_t$, where $(j,t) = (2,2), (3,3), (4,4)$.

Proof. This is Gritsenko–Nikulin [20], Section 5.1.1, cases $(t, II, \bar{0})$. The displayed sets are exactly $P_{t,II,\bar{0}}(\mathcal{M}_{t,II,\bar{0}})$ for $t = 2, 3, 4$. □

Lemma B.6 *(Product cone and chamber compatibility).* Let $t = 2, 3, 4$. If

$$f_t(nm, l) \neq 0, \qquad (n, l, m) \in \Gamma_t^{\mathrm{prod}},$$

then

$$(\alpha_t(n, l, m), \rho_t) \leq 0.$$

For $t = 2, 3$ the inequality is strict. For $t = 4$, equality occurs only on the parabolic ray

$$\alpha_t(n, l, m) \in \mathbb{Q}_{>0}\rho_4.$$

Proof. The Jacobi support satisfies

$$4tnm - l^2 \geq -1.$$

Since

$$(\alpha_t(n, l, m), \rho_t) = -2n + l - 2m,$$

the chamber inequality is $l \leq 2n + 2m$. For $t = 2, 3$, Gritsenko– Nikulin Lemma 2.5 gives the required strict chamber condition on the active support. For $t = 4$, the proof of Gritsenko–Nikulin (1.33) gives the same condition, with equality only in the parabolic case. Equivalently, equality and $4tnm - l^2 \geq -1$ force

$$n = m, \qquad l = 4n,$$

so

$$\alpha_4(n, 4n, n) = 8n\rho_4.$$

□

Theorem B.7 *(Gritsenko–Nikulin denominator rows).* Let $t = 2, 3, 4$ and put

t	$F(t)$	ϕ_t
2	Δ_2	$\phi_{0,2}$
3	Δ_1	$\phi_{0,3}$
4	$\Delta_{1/2}$	$\phi_{0,4}$

Let

$$\mathfrak{g}_t^{\mathrm{GN}} \quad (t = 2, 3, 4)$$

be the generalized Borcherds–Kac–Moody superalgebra whose Lorentzian root lattice is $\mathcal{M}_t$, whose Weyl group is $W_t = W^{(2)}(\mathcal{M}_t)$, whose Weyl vector is

$$\rho_t = \left(\frac{1}{2t}, \frac{1}{2}, \frac{1}{2t}\right),$$

and whose simple real roots are $P_t = P_{t,II,\bar{0}}$ from Proposition B.5. Let $\mathcal{M}_t$ denote the corresponding chamber. Then

$$P_{t,\bar{1}} = \varnothing, \qquad P_{t,\bar{0}} = P_t,$$

so every simple real root is even and $\epsilon_t(s_\delta) = -1$ for every $\delta \in P_t$; equivalently $\epsilon_t(w) = \det(w)$ on W_t.

Let

$$\phi_t(\tau, z) = \sum f_t(n, l) q^n r^l.$$

The integers $N_t(a)$ are the additive-side coefficients in Gritsenko–Nikulin (5.1), equivalently the signed supermultiplicities of imaginary simple roots. They are not the product exponents $f_t(nm, l)$. In the Gritsenko–Nikulin exponential normalization $e_t^{(n,l,m)} = q^n r^l s^{tm}$, the Weyl–Kac–Borcherds denominator identity is

$$\begin{aligned}\operatorname{den}(\mathfrak{g}_t^{\mathrm{GN}}) &= \sum_{w\in W_t} \epsilon_t(w)\left(e_t^{w(\rho_t)} + \sum_{a\in\mathbb{R}_{++}\mathcal{M}_t\cap M_t} N_t(a) e_t^{w(\rho_t+a)}\right)\\ &= e_t^{\rho_t} \prod_{(n,l,m)\in\Gamma_t^{\mathrm{prod}}} \left(1 - e_t^{(n,l,m)}\right)^{f_t(nm,l)} = F(t)(Z).\end{aligned}$$

Equivalently,

$$\operatorname{den}(\mathfrak{g}_2^{\mathrm{GN}}) = \Delta_2(Z), \qquad \operatorname{den}(\mathfrak{g}_3^{\mathrm{GN}}) = \Delta_1(Z), \qquad \operatorname{den}(\mathfrak{g}_4^{\mathrm{GN}}) = \Delta_{1/2}(Z).$$

Thus, for rows $j = 2, 3, 4$,

$$C'_j = 1, \qquad a_j = 1, \qquad L_j = M_j, \qquad \alpha_j = \alpha_j^{\mathrm{GN}}$$

in this normalization. The imaginary root spaces are determined here only as signed multiplicity spaces:

$$\operatorname{sdim}(\mathfrak{g}_{t,\alpha_t(n,l,m)}) = f_t(nm, l)$$

on the product side, with imaginary simple-root superdimensions $N_t(a)$ on the Weyl-sum side.

Proof. Gritsenko–Nikulin [20, Section 5.1] state that almost all identities proved earlier in their paper give denominator formulae for Lorentzian Kac–Moody algebras. Their formula (5.1) is exactly the displayed Weyl–Kac–Borcherds identity on $M_t = U(4t) \oplus \langle 2\rangle$, with $e^{(n,l,m)} = q^n r^l s^{tm}$.

For the cases $(t, II, \bar{o})$, $t = 2, 3, 4$, they give

$$W_{t,II,\bar{o}} = W^{(2)}(M_t), \qquad \rho_{t,II,\bar{o}} = \left(\frac{1}{2t}, \frac{1}{2}, \frac{1}{2t}\right),$$

and

$$P_{t,II,\bar{o}}(\mathcal{M}_{t,II,\bar{o}}) = P_{t,II,\bar{o}}(\mathcal{M}_{t,II,\bar{o}})_{\bar{o}}.$$

Hence the odd simple-real-root set is empty and the Weyl character is the ordinary reflection sign on the simple real reflections. The roots in P_t have square 2, and the generalized Cartan matrix is $((\delta, \delta'))_{\delta,\delta'\in P_t}$; for $t = 4$ this is the infinite symmetric parabolic matrix recorded by Gritsenko–Nikulin. For real roots $4tn - l^2 = -1$, the relevant Jacobi coefficient is 1: this is Gritsenko–Nikulin Lemma 2.5 for $t = 2, 3$ and the proof of Gritsenko–Nikulin (1.33) for $t = 4$.

The product factors are the full-paramodular products [20, (2.21), (2.20), (2.11)], already displayed in Proposition B.4. Their prefactors are exactly $e_t^{\rho_t}$. Therefore the denominator functions are Δ_2, Δ_1, and $\Delta_{1/2}$ with no further scalar or argument rescaling in the e_t normalization. □

PROPOSITION B.8 *(Congruence products).* For $j = 5, 6, 7, 8$ Clery–Gritsenko construct the following Borcherds products [18]. The weak Jacobi inputs are

j	ϕ_j	ρ_j^{pref}
5	$4(\xi_{1,0}^{(2)})^2 \in J_{0,1}^w(\Gamma_0(2))$	$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$
6	$3\xi_{3,1}^{(6)}\xi_{3,5}^{(6)} \in J_{0,1}^w(\Gamma_0(3))$	$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$
7	$2\xi_{2,1}^{(4)}\xi_{2,3}^{(4)} \in J_{0,1}^w(\Gamma_0(4))$	$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$
8	$2\xi_{1,0}^{(2)}(\tau, 2z) \in J_{0,2}^w(\Gamma_0(2))$	$(\frac{1}{4}, \frac{1}{2}, \frac{1}{2})$

where ρ_j^{pref} is read from the literal product prefactor $q^{\rho_\tau} r^{\rho_z} s^{\rho_\omega}$. At this stage it is a prefactor exponent, not a proved Weyl vector. In particular, for F_8 the M_2 product coordinate corresponding to $s^{1/2}$ is $(1/4, 1/2, 1/4)$ because $e^{\alpha_2(a,b,c)} = q^a r^b s^{2c}$. More precisely, if

$$S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right),$$

and

$$\phi_5 = \sum c_5(n,l) q^n r^l, \qquad \phi_5|_{0,1} S = \sum c_{5,S}(n,l) q^n r^l,$$

then

$$\begin{aligned} F_5(Z) &= \nabla_3(Z) \\ &= q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_1^{\mathrm{prod}}} (1 - q^n r^l s^m)^{c_5(nm,l)} (1 - q^{2n} r^{2l} s^{2m})^{c_{5,S}(nm,l)}. \end{aligned}$$

If

$$\phi_6 = \sum c_6(n,l) q^n r^l, \qquad \phi_6|_{0,1} S = \sum c_{6,S}(n,l) q^n r^l,$$

then

$$\begin{aligned} F_6(Z) &= \nabla_2(Z) \\ &= q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_1^{\mathrm{prod}}} (1 - q^n r^l s^m)^{c_6(nm,l)} (1 - q^{3n} r^{3l} s^{3m})^{c_{6,S}(nm,l)}. \end{aligned}$$

If

$$\phi_7 = \sum c_7(n,l) q^n r^l, \qquad \phi_7|_{0,1} S = \sum c_{7,S}(n,l) q^n r^l, \qquad \phi_7|_{0,1} M = \sum c_{7,M}(n,l) q^n r^l,$$

where $M = \left(\begin{smallmatrix} 1 & -1 \\ 2 & -1 \end{smallmatrix}\right)$, then

$$\begin{aligned} F_7(Z) &= \nabla_{3/2}(Z) \\ &= q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_1^{\mathrm{prod}}} (1 - q^n r^l s^m)^{c_7(nm,l)} \\ &\qquad \cdot (1 - q^{2n} r^{2l} s^{2m})^{c_{7,M}(nm,l)/2} (1 - q^{4n} r^{4l} s^{4m})^{c_{7,S}(nm,l)}. \end{aligned}$$

If

$$\phi_8 = \sum c_8(n,l) q^n r^l, \qquad \phi_8|_{0,2} S = \sum c_{8,S}(n,l) q^n r^l,$$

then

$$\begin{aligned} F_8(Z) &= Q_1(Z) \\ &= q^{1/4} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_2^{\mathrm{prod}}} (1 - q^n r^l s^{2m})^{c_8(nm,l)} (1 - q^{2n} r^{2l} s^{4m})^{c_{8,S}(nm,l)}. \end{aligned}$$

These four displays pin the automorphic products. By themselves they do not construct a BKM chamber, parity function, or Weyl-sum identity.

Proof. The formulae are Clery–Gritsenko [18, Theorem 3.1] and the four product displays for ∇_3, ∇_2, $\nabla_{3/2}$, and Q_1 at the end of Section 3. The second factors in F_5, F_6, F_8 and the second and third factors in F_7 come from the non-infinity cusps of $\Gamma_0(N)$ in their congruence Borcherds product theorem. □

PROPOSITION B.9 *(Product degree lattices for F_5, F_6, F_7, F_8).* For the product rows $j = 5, 6, 7, 8$ set

$$(t_5, N_5) = (1, 2), \qquad (t_6, N_6) = (1, 3), \qquad (t_7, N_7) = (1, 4), \qquad (t_8, N_8) = (2, 2).$$

The Clery–Gritsenko modular host is the orthogonal lattice

$$O_j = U \oplus U(N_j) \oplus \langle -2t_j \rangle$$

of signature $(2, 3)$. The single product-degree lattice for the monomial denominator is

$$L_j^{\mathrm{prod}} := M_{t_j}.$$

Define the cusp-labelled degree maps by

$$\alpha_5^{\mathrm{prod}}(n, l, m; \infty) = \alpha_1(n, l, m), \qquad \alpha_5^{\mathrm{prod}}(n, l, m; S) = \alpha_1(2n, 2l, 2m),$$
$$\alpha_6^{\mathrm{prod}}(n, l, m; \infty) = \alpha_1(n, l, m), \qquad \alpha_6^{\mathrm{prod}}(n, l, m; S) = \alpha_1(3n, 3l, 3m),$$
$$\alpha_7^{\mathrm{prod}}(n, l, m; \infty) = \alpha_1(n, l, m), \qquad \alpha_7^{\mathrm{prod}}(n, l, m; M) = \alpha_1(2n, 2l, 2m),$$
$$\alpha_7^{\mathrm{prod}}(n, l, m; S) = \alpha_1(4n, 4l, 4m),$$

and

$$\alpha_8^{\mathrm{prod}}(n, l, m; \infty) = \alpha_2(n, l, m), \qquad \alpha_8^{\mathrm{prod}}(n, l, m; S) = \alpha_2(2n, 2l, 2m).$$

The corresponding product-prefactor degrees are

$$\rho_5^{\mathrm{prod}} = \rho_6^{\mathrm{prod}} = \rho_7^{\mathrm{prod}} = \alpha_1(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), \qquad \rho_8^{\mathrm{prod}} = \alpha_2(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}).$$

For $\lambda \in L_j^{\mathrm{prod}}$ let $\mu_j(\lambda)$ be the sum of all cusp coefficients whose labelled degree is λ:

$$\mu_j(\lambda) = \sum_{\alpha_j^{\mathrm{prod}}(n,l,m;\kappa)=\lambda} \widetilde{c}_{j,\kappa}(nm, l),$$

where

$$\widetilde{c}_{j,\infty} = c_j, \qquad \widetilde{c}_{5,S} = c_{5,S}, \qquad \widetilde{c}_{6,S} = c_{6,S},$$
$$\widetilde{c}_{7,M} = \frac{1}{2} c_{7,M}, \qquad \widetilde{c}_{7,S} = c_{7,S}, \qquad \widetilde{c}_{8,S} = c_{8,S}.$$

The Clery–Gritsenko product is an automorphic product with integral exponents on its multiplier cover; in particular the row-7 exponents $\frac{1}{2} c_{7,M}(n, l)$ are integers on the displayed support. Let

$$\mathcal{K}_5 = \mathcal{K}_6 = \mathcal{K}_8 = \{\infty, S\}, \qquad \mathcal{K}_7 = \{\infty, M, S\},$$

and

$$\mathcal{D}_j := \Gamma_{t_j}^{\mathrm{prod}} \times \mathcal{K}_j.$$

The cusp-labelled product datum is

$$\mathcal{E}_j^{\mathrm{prod}} := (\mathcal{D}_j, \alpha_j^{\mathrm{prod}}, \mu_j).$$

Then

$$F_j(Z) = e^{\rho_j^{\mathrm{prod}}} \prod_{\lambda \in \alpha_j^{\mathrm{prod}}(\mathcal{D}_j)} \left(1 - e^{\lambda}\right)^{\mu_j(\lambda)} \qquad (j = 5, 6, 7, 8),$$

where $e^{\alpha_1(a,b,c)} = q^a r^b s^c$ and $e^{\alpha_2(a,b,c)} = q^a r^b s^{2c}$.

Proof. The displayed maps are the unique monomial-degree maps forced by the three products in Proposition B.8. For F_5 the non-infinity cusp contributes the monomials $q^{2n}r^{2l}s^{2m}$, hence degree $\alpha_1(2n, 2l, 2m)$. For F_6 the corresponding monomial is $q^{3n}r^{3l}s^{3m}$, hence degree $\alpha_1(3n, 3l, 3m)$. For F_7 the two non-infinity factors have monomials $q^{2n}r^{2l}s^{2m}$ and $q^{4n}r^{4l}s^{4m}$; the first has half exponent $c_{7,M}/2$ exactly as in the Clery–Gritsenko product. For F_8 the product has index $t = 2$: $e^{\alpha_2(n,l,m)} = q^n r^l s^{2m}$ and the non-infinity cusp contributes $e^{\alpha_2(2n,2l,2m)} = q^{2n}r^{2l}s^{4m}$. If two labelled cusp contributions land on the same lattice degree, exponents add; this is precisely the definition of μ_j.

Clery–Gritsenko [18, Theorem 3.1] realize these products as orthogonal Borcherds products for $U \oplus U(N_j) \oplus \langle -2t_j \rangle$. That theorem supplies the automorphic product and the divisor. It does not supply a Weyl chamber, simple real roots, a parity function, or a Weyl-sum identity on M_{t_j}. Thus L_j^{prod} and α_j^{prod} are product data, not a completed BKM denominator datum. The completed denominator datum for F_5, F_6, F_7, F_8 comes from the CHL theorem in Theorem B.13. □

Definition B.10 *(Formal product algebra of a row).* For a row j with product-degree function μ_j, let

$$\mathcal{V}_j^{\mathrm{prod}} = \bigoplus_{\lambda \in L_j^{\mathrm{prod}},\, \lambda > 0} \mathcal{V}_{j,\lambda}$$

be any graded super vector space with finite-dimensional homogeneous pieces and

$$\operatorname{sdim} \mathcal{V}_{j,\lambda} = \mu_j(\lambda).$$

Choose a Grothendieck representative: the even part has dimension $\mu_j(\lambda)$ when $\mu_j(\lambda) > 0$, the odd part of dimension $-\mu_j(\lambda)$ when $\mu_j(\lambda) < 0$, and the zero space when $\mu_j(\lambda) = 0$. Give it the abelian Lie superalgebra bracket:

$$[\mathcal{V}_{j,\lambda}, \mathcal{V}_{j,\lambda'}] = 0.$$

Its formal current envelope on E is

$$\mathsf{Prod}_{j,E} := U_E^{\mathrm{ch}}\Big(\mathcal{V}_j^{\mathrm{prod}} \otimes \mathcal{O}_E\Big).$$

Theorem B.11 *(Eight formal current rows).* Every Clery–Gritsenko diagonal-divisor row has a formal determinant object. For $j = 1$ this is $\mathsf{Den}_{\Delta_5,E}$ from Theorem 8.4. For $j = 2, 3, 4$ it is the current envelope of the Gritsenko–Nikulin BKM algebra $\mathfrak{g}_j^{\mathrm{GN}}$. For $j = 5, 6, 7, 8$ it is the formal product algebra $\mathsf{Prod}_{j,E}$. For $j = 5, 6, 7, 8$, Theorem B.13 also supplies a nonabelian CHL/WKB denominator algebra; the abelian product algebra in this theorem is the part constructed directly from the Clery–Gritsenko product exponents. For $j = 5, 6, 7, 8$ this is only the tautological virtual product object realizing the Clery–Gritsenko exponents. Its Beilinson–Drinfeld envelope is abelian; it carries no Weyl-sum side, no BKM bracket, and no physical protected BPS interpretation.

In all eight cases the formal determinant is the corresponding monic automorphic product:

$$e^{\rho_j} \operatorname{sdet}_{\mathcal{V}_j}(1 - e) = D_j(Z),$$

where $D_1 = 64^{-1}\Delta_5$ and $D_j = F_j$ for the displayed Clery–Gritsenko product normalizations $j > 1$, with the normalizations stated in Propositions B.4, B.8, and B.9.

Proof. For $j = 1$, Theorem 8.2 gives the Borcherds denominator algebra and Theorem 8.4 gives the formal current envelope. For $j = 2, 3, 4$, Theorem B.7 gives the BKM brackets and their denominator identities.

For $j = 5, 6, 7, 8$, Proposition B.9 constructs the degree lattice, the positive product support, the prefactor, and the signed exponent μ_j. The super vector space $\mathcal{V}_j^{\mathrm{prod}}$ realizes these signed dimensions. The Fock determinant lemma gives

$$e^{\rho_j^{\mathrm{prod}}} \prod_{\lambda} (1 - e^{\lambda})^{\mathrm{sdim}\, \mathcal{V}_{j,\lambda}} = e^{\rho_j^{\mathrm{prod}}} \prod_{\lambda} (1 - e^{\lambda})^{\mu_j(\lambda)} = F_j(Z).$$

The formal current enveloping construction gives a chiral algebra on E from this abelian Lie superalgebra. It is not a compact BPS factorization algebra. □

PROPOSITION B.12 *(Nonabelian promotion test for F_5, F_6, F_7, F_8).* Suppose that one of the congruence product rows

$$j \in \{5, 6, 7, 8\}$$

is the denominator of a generalized Borcherds–Kac–Moody superalgebra on the product lattice L_j^{prod}. Then the following additional data must exist:

$$\Big(P_j, \mathcal{M}_j, W_j, \epsilon_j, \rho_j, N_j(a)\Big).$$

They must satisfy

$$W_j = \langle s_\delta \mid \delta \in P_j \rangle, \qquad \mathcal{M}_j = \{x \mid (x, \delta) \leq 0 \text{ for all } \delta \in P_j\},$$

$$(\rho_j, \delta) = -\frac{(\delta, \delta)}{2} \qquad (\delta \in P_j),$$

and the Weyl–Kac–Borcherds identity

$$\begin{aligned} F_j(Z) &= \sum_{w \in W_j} \epsilon_j(w) \left(e^{w(\rho_j)} + \sum_{a \in L_j^{\mathrm{prod}} \cap \mathbb{R}_{>0} \mathcal{M}_j} N_j(a) e^{w(\rho_j + a)} \right) \\ &= e^{\rho_j^{\mathrm{prod}}} \prod_{\lambda \in \alpha_j^{\mathrm{prod}}(\mathcal{D}_j)} \left(1 - e^{\lambda}\right)^{\mu_j(\lambda)}. \end{aligned}$$

In particular the product already fixes the signed root multiplicity function μ_j and the product-prefactor degree

$$\rho_5^{\mathrm{prod}} = \rho_6^{\mathrm{prod}} = \rho_7^{\mathrm{prod}} = \alpha_1(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), \qquad \rho_8^{\mathrm{prod}} = \alpha_2(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}),$$

but it does not fix $P_j, \mathcal{M}_j, W_j, \epsilon_j$, or the imaginary simple-root coefficients $N_j(a)$.

Proof. The equalities in Proposition B.9 construct only the monomial degree and the total exponent $\mu_j(\lambda)$. A denominator theorem has a second side: the Weyl alternating sum. That side requires a reflection chamber, a set of simple real roots, a reflection character, and the imaginary simple-root contribution.

The Gritsenko–Nikulin chambers for M_1 and M_2 do not follow by notation. The forms $\nabla_3, \nabla_2, \nabla_{3/2}$, and Q_1 are congruence products with non-infinity cusp factors, whereas Δ_5 and Δ_2 are full-paramodular

denominator forms. Clery–Gritsenko construct the automorphic product, character, and diagonal divisor on

$$U \oplus U(N_j) \oplus \langle -2t_j \rangle,$$

but their product theorem does not identify the simple real roots or prove a Weyl anti-invariant additive expansion on M_{t_j}. Therefore the displayed six-tuple is an additional theorem to be proved, not a consequence of the product formula. □

Theorem B.13 *(Govindarajan CHL/WKB denominator rows).* In the CHL/twisted-dyon normalization of Cheng–Dabholkar and Govindarajan, the rows

$$F_5 = \nabla_3, \qquad F_6 = \nabla_2, \qquad F_7 = \nabla_{3/2}, \qquad F_8 = Q_1$$

are Weyl–Kac–Borcherds denominator forms. The row F_7 is taken on the μ_4-multiplier cover determined by the Clery–Gritsenko order-four multiplier. Thus F_7 is a denominator on the μ_4-cover, with no Vol. III row map, metaplectic boundary theorem, or CY/DT/chiral host. In the product-degree notation of this appendix the real-root data are

$$L_5 = L_6 = L_7 = M_1, \qquad P_5 = P_6 = P_7 = P^{(1)}, \qquad \rho_5 = \rho_6 = \rho_7 = \alpha_1(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}),$$

where

$$P^{(1)} = \{(0,-1,0),\ (1,1,0),\ (0,1,1)\},$$

and

$$L_8 = M_2, \qquad P_8 = P^{(2)}, \qquad \rho_8 = \alpha_2(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}),$$

where

$$P^{(2)} = \{(0,-1,0),\ (1,1,0),\ (1,3,1),\ (0,1,1)\}.$$

The Weyl chambers and Weyl groups are

$$\mathcal{M}_j = \{x \in L_j \otimes \mathbb{R} \mid (x,\delta) \le 0 \text{ for every } \delta \in P_j\}, \qquad W_j = \langle s_\delta \mid \delta \in P_j \rangle.$$

Every simple real root is even and the reflection character is

$$\epsilon_j(w) = \det(w).$$

For $j = 5, 6, 7, 8$ the denominator product side is expressed through the cusp-labelled product datum $\mathscr{E}_j^{\mathrm{prod}}$, with root multiplicity function μ_j. The Clery–Gritsenko character ν_j, restricted to the embedded CHL Weyl group W_j, is the reflection character:

$$\nu_j|_{W_j} = \epsilon_j = \det.$$

This restriction is part of the Govindarajan WKB denominator theorem, not a consequence of the Clery–Gritsenko product alone. The Cartan matrices are

$$A^{(1)} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix},$$

for F_5, F_6, F_7, and

$$A^{(2)} = \begin{pmatrix} 2 & -2 & -6 & -2 \\ -2 & 2 & -2 & -6 \\ -6 & -2 & 2 & -2 \\ -2 & -6 & -2 & 2 \end{pmatrix},$$

for F_8.

Equivalently, for $x = (u, v, w)$, the two chambers are

$$\mathcal{M}_1 = \{x \mid v \geq 0,\ v \leq 2w,\ v \leq 2u\},$$

and

$$\mathcal{M}_2 = \{x \mid v \geq 0,\ v \leq 4w,\ 3v \leq 4(u + w),\ v \leq 4u\}.$$

There are integral imaginary simple-root coefficients $N_j(a)$, depending on the CHL twist, such that

$$\begin{aligned} F_j(Z) &= \sum_{w \in W_j} \det(w) \left(e^{w(\rho_j)} + \sum_{a \in L_j \cap \mathbb{R}_{>0} \mathcal{M}_j} N_j(a) e^{w(\rho_j + a)} \right) \\ &= e^{\rho_j} \prod_{\lambda \in \alpha_j^{\mathrm{prod}}(\mathcal{D}_j)} \left(1 - e^{\lambda}\right)^{\mu_j(\lambda)} \qquad (j = 5, 6, 7, 8). \end{aligned}$$

Thus the six-tuples

$$(P_j, \mathcal{M}_j, W_j, \epsilon_j, \rho_j, N_j(a))$$

are constructed for $j = 5, 6, 7, 8$ by the CHL denominator theorem. They are not consequences of the Clery–Gritsenko product formula alone. In the monic Clery–Gritsenko/Govindarajan product variables of Proposition B.8, the identities have

$$C'_j = 1, \qquad a_j = 1 \qquad (j = 5, 6, 7, 8).$$

This is not the doubled $N = 1$ normalization of Lemma B.3.

Remark B.14 *(F_7 on the μ_4-multiplier cover).* The row $F_7 = \nabla_{3/2}$ is the Clery–Gritsenko diagonal-divisor form of type $(t, N) = (1, 4)$, weight $3/2$, on $\Gamma_0^{(2)}(4)$ with an order-four multiplier. After pullback to the associated μ_4-multiplier cover, Govindarajan's CHL/WKB theorem identifies it with the Weyl–Kac–Borcherds denominator of $\mathcal{G}_1(4)$. It is the μ_4-cover denominator identity, with no Vol. III physical $\Phi_7^{\mathrm{V}_3}$, no metaplectic boundary host, and no CY/DT or chiral/Hall theorem.

The square identity $F_7^2 = \Phi_3$ is only a scalar square identity. It does not carry the reflection sign, parity, Weyl chamber, or Weyl-alternating side. The coefficient convention for the additive side is fixed in Lemma B.16.

LEMMA B.15 *(Coefficient normalisation for ∇_3 and ∇_2).* Let

$$\psi_5(\tau, z) = \vartheta(\tau, z)\eta(\tau)\eta(2\tau)^4 = \sum g_5(u, v) q^{u/2} r^{v/2},$$

and

$$\psi_6(\tau, z) = \vartheta(\tau, z)\eta(3\tau)^3 = \sum g_6(u, v) q^{u/2} r^{v/2}.$$

For $a = \alpha_1(A, B, C) \in M_1 \cap \mathbb{R}_{\geq 0}\mathcal{M}_1$, put

$$n = 2A + 1, \qquad l = 2B + 1, \qquad m = 2C + 1.$$

In the plus-sign convention of Theorem B.13, the imaginary simple-root coefficients for the rows F_5 and F_6 are

$$N_5(a) = \sum_{d|(n,l,m)} d^2\, g_5\left(\frac{nm}{d^2}, \frac{l}{d}\right),$$

and

$$N_6(a) = \sum_{d|(n,l,m)} \left(\frac{-3}{d}\right) d\, g_6\left(\frac{nm}{d^2}, \frac{l}{d}\right).$$

Govindarajan–Gopala Krishna write the same coefficients with the opposite sign in the imaginary-root correction term:

$$m_j(a) = -N_j(a) \qquad (j = 5, 6).$$

Proof. Govindarajan–Gopala Krishna construct the additive lift whose Weyl alternating side gives the denominator identity for the corresponding CHL square roots [15, (4.2), (4.4), (4.5), (4.8)]. Their Maass lift gives the displayed divisor sums from the Jacobi seeds ψ_5 and ψ_6, and their Weyl–Kac–Borcherds sum writes these coefficients as the imaginary simple-root correction. The sign difference is only the convention:

$$e^\rho + \sum_a N_j(a) e^{\rho + a},$$

whereas [15] writes the correction with a minus sign. □

Lemma B.16 *(Coefficient normalisation for $\nabla_{3/2}$).* Let

$$\nabla_{3/2}(Z) = \Delta^{(4)}_{3/2}(Z) = \frac{1}{8}\theta_5(2Z)\theta_7(2Z)\theta_9(2Z) = \sum_{n,l,m} A_7(n, l, m) q^{n/2} r^{l/2} s^{m/2},$$

with $A_7(1, 1, 1) = 1$. The first term is

$$\left(\sqrt{r} - \frac{1}{\sqrt{r}}\right)\sqrt{q}\,\sqrt{s} + \cdots,$$

so $A_7(1, -1, 1) = -1$. For $a = \alpha_1(A, B, C) \in M_1 \cap \mathbb{R}_{\geq 0}\mathcal{M}_1$, put

$$n = 2A + 1, \qquad l = 2B + 1, \qquad m = 2C + 1.$$

Then the imaginary simple-root coefficient in the plus-sign convention of Theorem B.13 is

$$N_7(\alpha_1(A, B, C)) = A_7(2A + 1, 2B + 1, 2C + 1) = [q^{A+1/2} r^{B+1/2} s^{C+1/2}]\nabla_{3/2}(Z).$$

In Govindarajan–Gopala Krishna's WKB sign convention,

$$m_7(a) = -N_7(a).$$

No Maass divisor-sum formula for N_7 is asserted here.

Proof. Clery–Gritsenko construct $\nabla_{3/2}$ as the order-four multiplier row. Govindarajan–Gopala Krishna identify the square-root theta product and the square relation

$$F_7^2 = \Phi_3$$

in the composite $N = 4$ CHL case [16]. Factoring the Weyl vector gives

$$e^{\rho_7 + a} = q^{A+1/2} r^{B+1/2} s^{C+1/2},$$

which is exactly the displayed coefficient extraction. The sourced divisor-sum formula belongs to the square F_7^2, with its composite $N = 4$ Jacobi seed. A direct prime-type half-index formula for F_7 introduces fractional divisor powers at non-primitive chamber points, whereas the theta product has integral coefficients. □

LEMMA B.17 *(Q_1 in M_2 notation).* In M_2,

$$e^{\alpha_2(n,l,m)} = q^n r^l s^{2m}, \qquad (\alpha_2(n,l,m), \alpha_2(n',l',m')) = 2ll' - 8(nm' + n'm).$$

The four simple real roots of the Q_1 row are

$$\delta_1 = (0,-1,0), \quad \delta_2 = (1,1,0), \quad \delta_3 = (1,3,1), \quad \delta_4 = (0,1,1),$$

with Gram matrix $A^{(2)}$. The Weyl vector is

$$\rho_8 = \alpha_2(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}), \qquad e^{\rho_8} = q^{1/4} r^{1/2} s^{1/2},$$

and

$$(\rho_8, \delta_i) = -1 \qquad (1 \le i \le 4).$$

Let $A_8(n,l,m)$ be defined by

$$Q_1(Z) = \sum_{n,l,m} A_8(n,l,m) q^{n+1/4} r^{l+1/2} s^{2m+1/2}.$$

Govindarajan's theta-product formula, in his characteristic notation, is

$$Q_1(Z) = \frac{1}{4} \vartheta \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}(Z') \vartheta \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}(Z'), \qquad Z' = \begin{pmatrix} \tau & 2z \\ 2z & 4\sigma \end{pmatrix}.$$

Equivalently,

$$A_8(n,l,m) = \frac{1}{4} \sum_{\mathcal{S}(n,l,m)} (-1)^{j+k+h},$$

where $\mathcal{S}(n,l,m)$ is the set of $(i,j,k,h) \in \mathbb{Z}^4$ satisfying

$$\frac{1}{2}\left((i+\tfrac{1}{2})^2 + (k+\tfrac{1}{2})^2\right) = n + \tfrac{1}{4},$$

$$2(i+\tfrac{1}{2})j + 2(k+\tfrac{1}{2})(h+\tfrac{1}{2}) = l + \tfrac{1}{2},$$

and

$$2j^2 + 2(h+\tfrac{1}{2})^2 = 2m + \tfrac{1}{2}.$$

Then the imaginary simple-root coefficient in the plus-sign convention is

$$N_8(\alpha_2(n,l,m)) = A_8(n,l,m)$$

for the chamber-supported imaginary shifts. Govindarajan's expansion has coefficient -1 at the monomials

$$r^{-1}, \qquad qr, \qquad rs^2, \qquad qr^3s^2.$$

After factoring e^{ρ_8} these are precisely the four simple real root monomials for

$$\delta_1, \quad \delta_2, \quad \delta_4, \quad \delta_3,$$

so the signed real-simple-root supermultiplicity is $+1$ in each case. Their even parity and multiplicity-one real-root data are the Cheng–Dabholkar/Govindarajan CHL denominator data.

Proof. The first display is the definition of α_2 and the Gram matrix on M_2. The simple roots and Weyl vector are Cheng–Dabholkar's $N = 2$ chamber data translated through

$$\alpha_2(n,l,m) \longleftrightarrow \begin{pmatrix} 2n & l \\ l & 4m \end{pmatrix}.$$

Govindarajan identifies Q_1 with the row $\mathcal{G}_2(4)$ and gives both the product formula and the first Fourier terms. The order-four Clery–Gritsenko character restricts to the Weyl sign on the reflection group by the oddness under the elementary reflection $z \mapsto -z$ and the chamber symmetries used in the CHL proof. □

Proof of Theorem B.13. Cheng–Dabholkar identify, for CHL level $N = 1, 2, 3$, the Weyl group, simple real roots, Weyl chamber, and Weyl denominator identity of the dyon algebra [8, (2.10)–(2.16), (3.16), (3.22), (3.23), (3.26), (3.42), (5.1), (5.2), (5.8), (5.10), (5.12)–(5.15), Appendix A]. For $N = 1$ the simple real roots are $P^{(1)}$ and the Cartan matrix is $A^{(1)}$; for $N = 2$ the simple real roots are $P^{(2)}$ and the Cartan matrix is $A^{(2)}$. Govindarajan then treats twisted dyons and proves that, for fixed N, the algebras $\mathcal{G}_N(M)$ have the same real simple roots, Weyl vector, and Cartan matrix, while the imaginary simple roots depend on M [14, Section 4.2, Section 4.3, Table 3].

Govindarajan's Table 3 gives

$$\nabla_3 \longleftrightarrow \mathcal{G}_1(2), \qquad \nabla_2 \longleftrightarrow \mathcal{G}_1(3), \qquad \nabla_{3/2} \longleftrightarrow \mathcal{G}_1(4), \qquad Q_1 \longleftrightarrow \mathcal{G}_2(4).$$

Section 4.2 of [14] checks the required ingredients: sum and product representations, integrality, correct multiplicity of the simple real roots, and the sign change of the square root under a Weyl reflection. Section 4.3 states the Weyl–Kac–Borcherds denominator formula for $\mathcal{G}_N(M)$. For $F_7 = \nabla_{3/2}$ the theorem is on the μ_4-multiplier cover, with no Vol. III boundary partition identity and no CY/DT/chiral host. For Q_1, Govindarajan supplies the WKB denominator data [14, Appendix C, Appendix D]. The dyon-counting additive-seed issue for the $(2,4)$ and $(4,2)$ examples plays no role. The later hyperbolic and umbral extensions of [17] are also outside this proof: for $N = 5, 6$ they replace W by W_+, leave the sum side incompletely characterized, and treat some rows as product-only or candidate BKM data [17, Sections 3.3 and 4].

The real-root data translate to this notation as follows. On M_t,

$$((n,l,m),(n',l',m')) = 2ll' - 4t(nm' + n'm).$$

With $t = 1$, the set $P^{(1)}$ has Gram matrix $A^{(1)}$ and $\rho = \alpha_1(1/2, 1/2, 1/2)$ satisfies $(\rho, \delta) = -1$ for every $\delta \in P^{(1)}$. With $t = 2$, the set $P^{(2)}$ has Gram matrix $A^{(2)}$ and $\rho = \alpha_2(1/4, 1/2, 1/4)$ satisfies $(\rho, \delta) = -1$ for every $\delta \in P^{(2)}$. These are exactly the Weyl-vector equations $(\rho, \delta) = -(\delta, \delta)/2$ because all displayed real roots have square 2. □

Remark B.18 *(Nonabelian row data).* The row data are:

j	automorphic data $(F_j, \Gamma_j, \nu_j, \rho_j^{\mathrm{pref}})$	lattice and degree datum
1	constructed here	constructed here
2, 3, 4	GN denominator theorem	$L_j = M_j,\ \alpha_j = \alpha_j^{\mathrm{GN}}$
5, 6, 7	CG product; Govindarajan denominator	$P^{(1)}, \mathcal{M}_1, W_1, \epsilon = \det$ constructed
8	CG product; Govindarajan denominator	$P^{(2)}, \mathcal{M}_2, W_2, \epsilon = \det$ constructed

Thus all eight rows have formal product/current-envelope objects. Rows 1–4 have the Gritsenko–Nikulin denominator algebras. Rows 5–8 have the Govindarajan CHL/WKB denominator algebras in the CHL normalization. The row F_7 lives on the μ_4-multiplier cover and gives no Vol. III physical boundary theorem.

Open Problem B.19 *(Physical row map and boundary host).* Construct, for every physical row, a datum

$$(\iota_j, X_j, \theta_j, \mathcal{P}_j)$$

such that

$$\iota_j : \Phi_j^{\mathrm{phys}} \longleftrightarrow F_i^{\mathrm{CG}}, \qquad Z_{\mathrm{red}}^{X_j} = C_j (F_i^{\mathrm{CG}})^{-2},$$

and

$$\mathrm{Prim}_{\mathrm{prot}} \mathcal{P}_j \cong \mathfrak{g}_i.$$

Here θ_j fixes the modular group or cover, character, divisor, Jacobi seed, normalization, orientation, Behrend sign, and Fourier chamber. The CHL/WKB theorem proves the denominator identities for $F_5, \ldots, F_8$. The physical row map and the boundary host are the open data.

PROPOSITION B.20 *(Boundary weights).* In the primary Clery–Gritsenko enumeration,

$$\{j : \mathrm{wt}(F_j) \in \mathbb{Z}\} = \{1, 2, 3, 5, 6, 8\}, \qquad \{j : \mathrm{wt}(F_j) \notin \mathbb{Z}\} = \{4, 7\}.$$

The Vol. III comparison notation

$$\Phi_1^{\mathrm{V3}}, \ldots, \Phi_8^{\mathrm{V3}}, \qquad \mathrm{wt}(\Phi_j^{\mathrm{V3}}) = (5, 2, 1, 1, \tfrac{1}{2}, 1, \tfrac{1}{4}, 0),$$

does not define the Clery–Gritsenko row system and defines no physical atlas in this paper.

Proof. The Clery–Gritsenko theorem gives the exact weights in the first display. The comparison weights

$$(5, 2, 1, 1, \tfrac{1}{2}, 1, \tfrac{1}{4}, 0)$$

are not the primary-source row labels of [18]. The Borcea–Voisin route [4], [35] and reduced curve-counting route [32], [31], [7] require a separate construction of X_j, the reduced theory, and the product identity. No such construction is used in the $N = 1$ theorem. □

Remark B.21 *(Boundary rows).* The Clery–Gritsenko weights are

$$(5, 2, 1, \tfrac{1}{2}, 3, 2, \tfrac{3}{2}, 1),$$

and no Clery–Gritsenko row has weight $1/4$ or weight 0. Also $F_4 = \Delta_{1/2}$, so F_4 has half-integral weight. A physical $\Phi_4^{V_3}$ identification requires a row map. A boundary partition identity requires the row map, the reduced theory, and the duality signs.

Remark B.22 *(Vol. III row map).* The symbols F_i and $\Phi_j^{V_3}$ label different row systems. When a collision is possible, write F_i^{CG} for the Clery–Gritsenko rows in the present order, T_a^{CG} for the same catalogue in triple order, and Φ_j^{cmp} for the comparison symbols. The rows F_i^{CG} are the diagonal-divisor forms of [18]; in this paper's order their weights are

$$(5, 2, 1, \tfrac{1}{2}, 3, 2, \tfrac{3}{2}, 1).$$

In triple order the same Clery–Gritsenko catalogue is indexed by

$$(t, N) = (1, 1), (2, 1), (1, 2), (3, 1), (1, 3), (4, 1), (1, 4), (2, 2),$$

where the weights are

$$(5, 2, 3, 1, 2, \tfrac{1}{2}, \tfrac{3}{2}, 1).$$

The permutation from that triple order to the present F_i-order is

$$(1, 2, 5, 3, 6, 4, 7, 8).$$

The comparison symbols

$$\Phi_1^{cmp}, \dots, \Phi_8^{cmp}, \qquad \mathrm{wt}(\Phi_j^{cmp}) = (5, 2, 1, 1, \tfrac{1}{2}, 1, \tfrac{1}{4}, 0),$$

are therefore not Clery–Gritsenko row labels. A row map

$$\Phi_j^{cmp} \longleftrightarrow F_i$$

requires the modular group or cover, character, divisor, Jacobi seed at every cusp, product-degree lattice, Weyl vector, real simple roots, reflection character, and, on the physical side, the CY/DT/chiral host and reduced partition identity. This paper contains no such datum.

Consequently the direct identifications

$$\Phi_5^{cmp} = F_5, \qquad \Phi_7^{cmp} = F_7, \qquad \Phi_8^{cmp} = F_8$$

are false already by weight. The Clery–Gritsenko catalogue contains no weight-$1/4$ row and no weight-0 row. The CHL square identities, such as $\nabla_3^2 = \Phi_6$ and $\nabla_2^2 = \Phi_4$, are scalar square identities; they are not row maps and do not by themselves carry the denominator sign, parity, or Weyl alternating side.

Remark B.23 *(Boundary host).* A boundary identity

$$Z_j = C_j(\Phi_j^{V_3})^{-2}$$

requires more than a Clery–Gritsenko product, a fractional weight, or a formal current envelope. It requires a row map $\Phi_j^{V_3} \leftrightarrow F_i$ fixing the weight, group, multiplier, cover, divisor, Jacobi input, and

normalization; a Calabi–Yau or orbifold host X_j with reduced obstruction theory, orientation, Behrend sign, Fourier chamber, and a partition theorem

$$Z_{\mathrm{red}}^{X_j} = C_j F_i^{-2};$$

and a compact oriented Hall/chiral datum whose protected primitive bracket satisfies the Borcherds relations for the same row. In the absence of these data, the boundary rows carry only their automorphic product and any separately proved denominator theorem. They carry no CY/DT partition identity, no physical BPS factorization algebra, and no fractional chiral power.

The following comparison table is restricted to the $N = 1$ Igusa boundary. It contains no row map

$$\Phi_j^{\mathrm{V}_3} \longleftrightarrow F_i$$

and no physical row-map denominator theorem for any congruence row. The separate Govindarajan CHL/WKB denominator theorem remains Theorem B.13.

C $N = 1$ Vol. III Boundary Conditions

Let a Vol. III chambered Hall-descent datum satisfy the five comparison conditions of Theorem 8.8. Its $N = 1$ decategorified boundary is then constrained by the Igusa datum

$$\begin{aligned}\Gamma_{\mathrm{eff}} = \{(n,l,m) \in \mathbb{Z}^3 \mid m > 0,\ n \geq 0,\ l \in \mathbb{Z}\}\\ \cup \{(n,l,0) \in \mathbb{Z}^3 \mid n > 0,\ l \in \mathbb{Z}\}\\ \cup \{(0,l,0) \in \mathbb{Z}^3 \mid l < 0\} \subset \mathbb{Z}^3.\end{aligned}$$

Vol. III Hall descent	Igusa boundary
Gram index lattice	$\Gamma_{\mathrm{ind}} = \mathbb{Z}^3$
sector completion	Γ_{eff}
active denominator support	$\Gamma_{\mathrm{act}} = \{\gamma \in \Gamma_{\mathrm{eff}} \mid f(nm,l) \neq 0\}$
Lorentzian degree	$\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$
orientation character	$\epsilon_o = \nu_{\Delta_5}$
automorphic boundary	$\mathrm{Borch}(\phi_{0,1}) = \Delta_5$
BKM denominator	$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$
denominator algebra	$\mathcal{A}_{\mathrm{den}} = \mathfrak{g}_{\Delta_5}$
formal current envelope	$\mathsf{Den}_{\Delta_5,E} = U_E^{\mathrm{ch}}(\mathfrak{g}_{\Delta_5} \otimes \mathcal{O}_E)$
scalar square	$Z_{\mathrm{OP}}^X = Z_{\mathrm{DT}}^X = -4096\Delta_5^{-2}$ (proved).

The degree map satisfies

$$(\alpha(\gamma), \alpha(\eta)) = -2\langle\gamma,\eta\rangle_{\mathrm{ind}}, \qquad (\alpha(n,l,m), \alpha(n,l,m)) = -2(4nm - l^2).$$

The real walls are

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3.$$

The Weyl vector is

$$\rho = f_2 - \frac{1}{2}f_3 + f_{-2}, \qquad (\rho, \delta_i) = -1.$$

These equations give the denominator polarisation of the same Gram-index lattice whose cusp polarisation gives the Fock determinant. The cone is generated by the active support. Triples in Γ_{eff} with zero Jacobi coefficient fix ordering and contribute no visible product factors. The product alone does not exclude invisible zero-superdimension root spaces.

The comparison is the $N = 1$ normalization

$$\mathrm{Borch}(\phi_{0,1}) = \Delta_5, \qquad \mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z), \qquad \kappa_{\mathrm{BKM}}(\mathfrak{g}_{\Delta_5}) = \mathrm{wt}(\Delta_5) = 5.$$

It contains no denominator theorem beyond the $N = 1$ theorem and no row map

$$\Phi_j^{\mathrm{V}_3} \longleftrightarrow F_i.$$

The protected scalar has the form

$$Z_{\square}^{X} = C_{\square}\Delta_5^{-2}.$$

The Oberdieck–Pixton and reduced Hilbert-scheme DT branches have constant -4096 in the quotient-by-E normalization of Theorem 5.2. The chiral/denominator half is Δ_5.

Vol. III's compact Hall–Drinfeld double realizes a protected bracket from the compactification only when the five comparison data of Theorem 8.8 are supplied: finite-first oriented Hall descent, protected integration, charge additive Hall primitives, Borcherds relations inside the Hall super-commutator, and the completed Hopf pairing with orientation character ν_{Δ_5}. Homotopy descent must produce the positive half and its completion data. The Igusa boundary identifies the denominator algebra $\mathfrak{g}_{\Delta_5}$ and its protected decategorified multiplicities before those comparison data are proved.

For the Clery–Gritsenko rows F_5, F_6, F_7, F_8, this table contains no nonabelian chamber, no simple real roots, no reflection character, and no Weyl-sum side. Rows F_5, F_6, F_7, F_8 use the separate Govindarajan CHL/WKB denominator theorem when a nonabelian bracket is needed. For F_7 this is a theorem on the μ_4-multiplier cover determined by the Clery–Gritsenko order-four multiplier. None of these denominator rows supplies a Vol. III row map or a CY/DT/chiral boundary host.

Bibliography

[1] A. N. Andrianov, *Euler products corresponding to siegel modular forms of genus 2*, Russian Math. Surveys **29** (1974), no. 3, 45–116.

[2] K. Behrend, *Donaldson–Thomas type invariants via microlocal geometry*, Annals of Mathematics **170** (2009), 1307–1338.

[3] A. Beilinson and V. Drinfeld, *Chiral algebras*, American Mathematical Society, 2004.

[4] C. Borcea, *K3 surfaces with involution and mirror pairs of calabi-yau manifolds*, Mirror Symmetry II, AMS/IP Stud. Adv. Math. **1** (1997), 717–743.

[5] R. E. Borcherds, *Automorphic forms with singularities on grassmannians*, Inventiones Mathematicae **132** (1998), no. 3, 491–562, available at `alg-geom/9609022`.

[6] J. Bryan, *The Donaldson–Thomas theory of K3 x E via the topological vertex*, Geometry of Moduli **14** (2018), 35–64, available at `arXiv:1504.02920`.

[7] J. Bryan, G. Oberdieck, R. Pandharipande, and Q. Yin, *Curve counting on abelian surfaces and threefolds*, Algebraic Geometry **5** (2018), 398–463, available at `arXiv:1506.00841`.

[8] M. C. N. Cheng and A. Dabholkar, *Borcherds–Kac–Moody symmetry of $N = 4$ dyons*, Communications in Number Theory and Physics **3** (2009), no. 1, 59–110, available at `arXiv:0809.4258`.

[9] P. Deligne, *Valeurs de fonctions l et periodes d'integrales*, Proc. Sympos. Pure Math. **33** (1979), 313–346.

[10] R. Dijkgraaf, G. Moore, E. Verlinde, and H. Verlinde, *Elliptic genera of symmetric products and second quantized strings*, Communications in Mathematical Physics **185** (1997), 197–209, available at `hep-th/9608096`.

[11] R. Dijkgraaf, E. Verlinde, and H. Verlinde, *Counting dyons in $N = 4$ string theory*, Nuclear Physics B **484** (1997), 543–561, available at `hep-th/9607026`.

[12] M. Eichler and D. Zagier, *The theory of Jacobi forms*, Progress in Math. 55, Birkhauser, 1985.

[13] E. Freitag, *Siegelsche modulfunktionen*, Springer, 1983.

[14] S. Govindarajan, *BKM lie superalgebras from counting twisted CHL dyons*, Journal of High Energy Physics **2011** (2011), no. 5, 089, available at `arXiv:1006.3472`.

[15] S. Govindarajan and K. Gopala Krishna, *Generalized Kac–Moody algebras from CHL dyons*, Journal of High Energy Physics **2009** (2009), no. 4, 032, available at `arXiv:0807.4451`.

[16] ———, *BKM lie superalgebras from dyon spectra in $\mathbb{Z}_N$-CHL orbifolds for composite N*, Journal of High Energy Physics **2010** (2010), no. 5, 014, available at `arXiv:0907.1410`.

[17] S. Govindarajan and S. Samanta, *BKM lie superalgebras from counting twisted CHL dyons – II*, Nuclear Physics B **948** (2019), 114770, available at `arXiv:1905.06083`.

[18] V. Gritsenko and F. Cléry, *The Siegel modular forms of genus 2 with the simplest divisor*, Proceedings of the London Mathematical Society **102** (2011), no. 6, 1024–1052, available at `arXiv:0812.3962`.

[19] V. A. Gritsenko and V. V. Nikulin, *Siegel automorphic form corrections of some Lorentzian Kac–Moody lie algebras*, American Journal of Mathematics **119** (1997), no. 1, 181–224, available at `alg-geom/9504006`.

[20] ———, *Automorphic forms and Lorentzian Kac–Moody algebras. part II*, International Journal of Mathematics **9** (1998), no. 2, 201–275, available at `alg-geom/9611028`.

[21] J. A. Harvey and G. Moore, *Algebras, bps states, and strings*, Nuclear Physics B **463** (1996), 315–368, available at `hep-th/9510182`.

[22] V. G. Kac, *Vertex algebras for beginners*, American Mathematical Society, 1998.

[23] V.G. Kac, *Infinite dimensional lie algebras*, Cambridge University Press, 1990.

[24] W. Kohnen and D. Zagier, *Modular forms with rational periods*, Modular forms (Durham, 1983), Ellis Horwood (1984), 197–249.

[25] H. Maass, *Die multiplikatorsysteme zur siegelschen modulgruppe*, Nachrichten der Akad. der Wissen. Gottingen (II. Math. Phys.Klasse) Nr 11, 1964.

[26] Yu. I. Manin, *Parabolic points and zeta-functions of modular curves*, Izv. Akad. Nauk SSSR Ser. Mat. **36** (1972), 19–66.

[27] V.V. Nikulin, *Surfaces of type k3 with finite automorphism group and picard group of rank three*, Proc. Steklov. Math. Inst. 165, 1984.

[28] G. Oberdieck, *On reduced stable pair invariants*, Mathematische Zeitschrift **289** (2018), no. 1–2, 323–353, available at `arXiv:1605.04631`.

[29] G. Oberdieck and R. Pandharipande, *Curve counting on K3 x E, the Igusa cusp form χ_{10}, and descendent integration*, K3 surfaces and their moduli, 2016, pp. 245–278.

[30] G. Oberdieck and A. Pixton, *Holomorphic anomaly equations and the Igusa cusp form conjecture*, Inventiones Mathematicae **213** (2018), no. 2, 507–587, available at `arXiv:1706.10100`.

[31] J. Oh and R. P. Thomas, *Complex Kuranishi structures and counting sheaves on Calabi–Yau 4-folds, II* (2023), available at `arXiv:2305.16441`.

[32] ———, *Counting sheaves on Calabi–Yau 4-folds, I*, Duke Mathematical Journal **172** (2023), no. 7, 1333–1409, available at `arXiv:2009.05542`.

[33] R. P. Thomas, *A holomorphic Casson invariant for Calabi–Yau 3-folds, and bundles on K3 fibrations*, Journal of Differential Geometry **54** (2000), no. 2, 367–438, available at `math/9806111`.

[34] G. van der Geer, *Siegel modular forms*, arXiv, 2007.

[35] C. Voisin, *Miroirs et involutions sur les surfaces k3*, Journees de geometrie algebrique d'Orsay, Asterisque **218** (1993), 273–323.

[36] D. Zagier, *Sur la conjecture de Saito–Kurokawa (d'après H. Maass)*, Séminaire Bourbaki 1979/80 **842** (1981), 371–394.